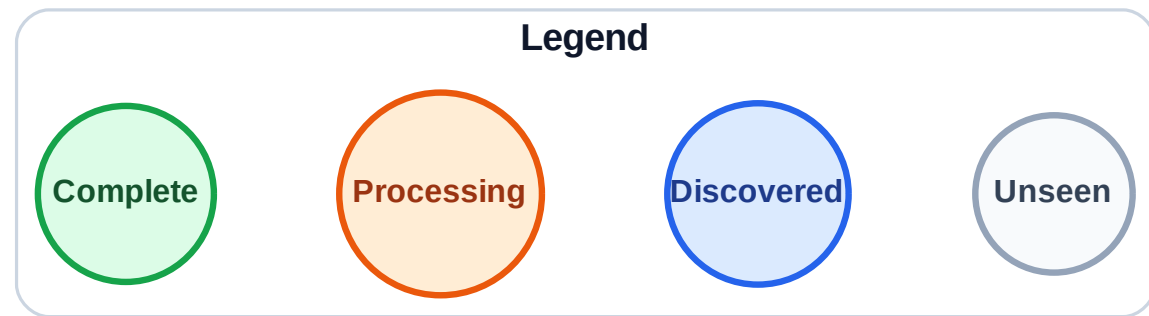
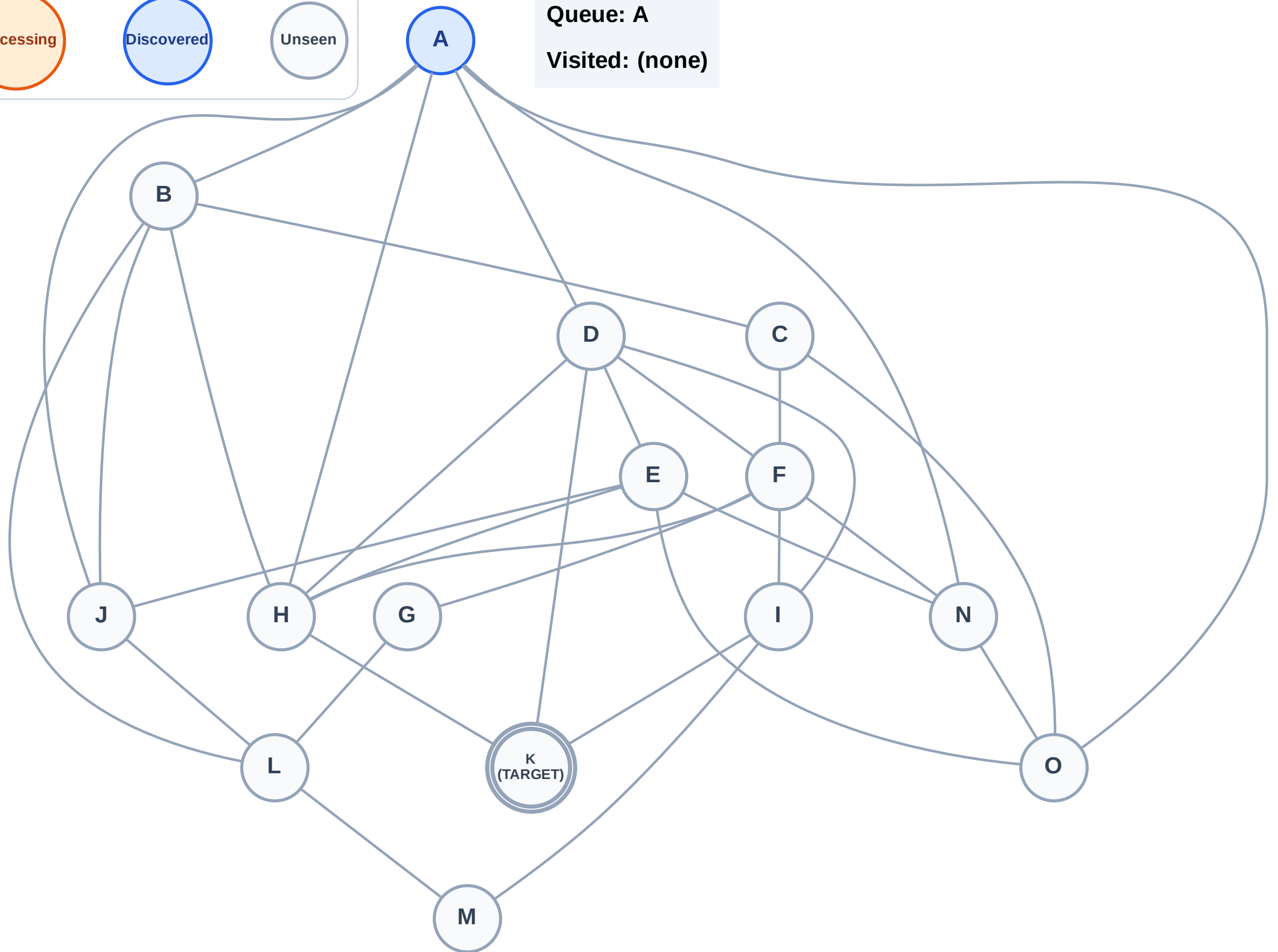


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

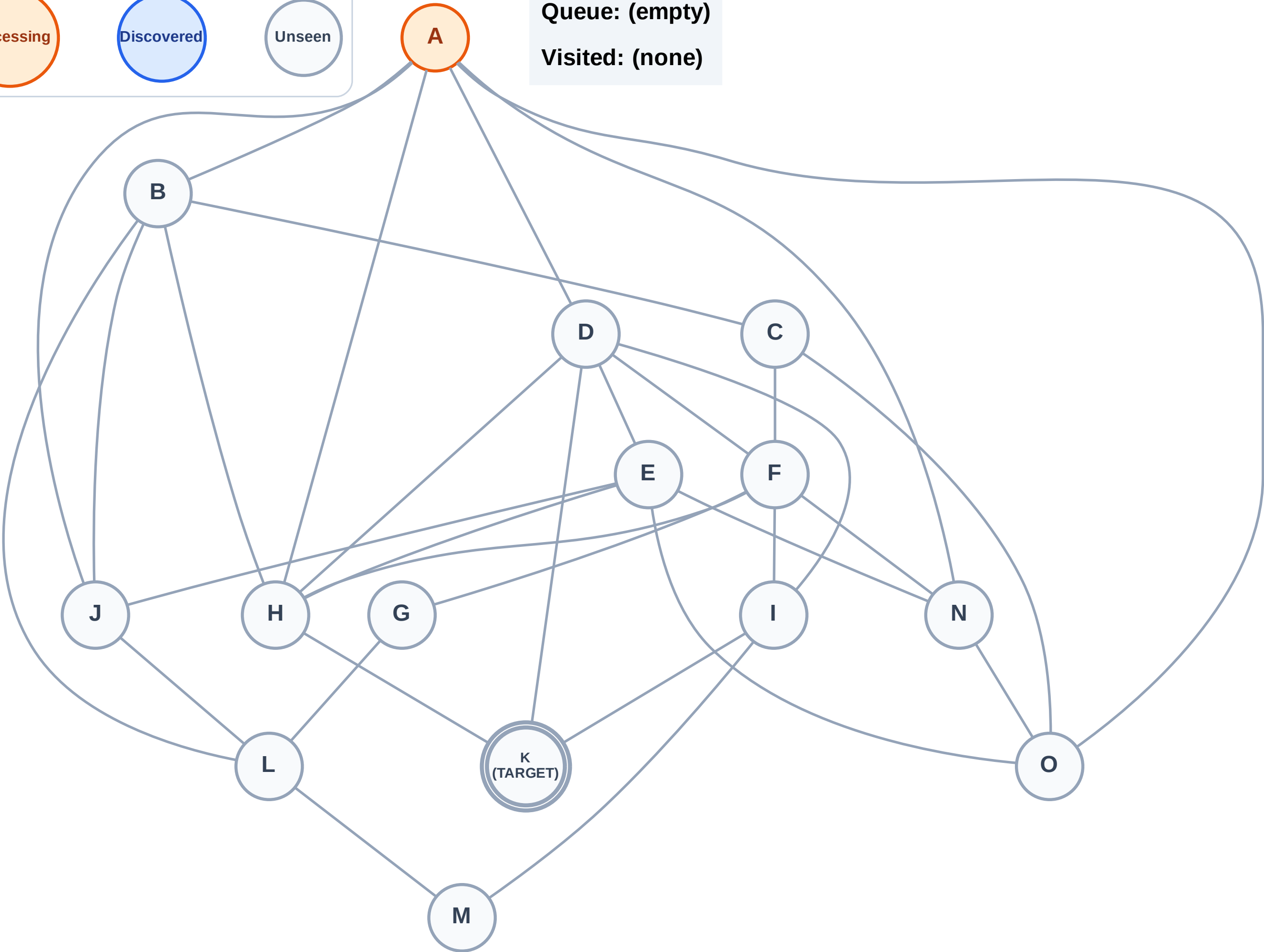
Processing

Discovered

Unseen

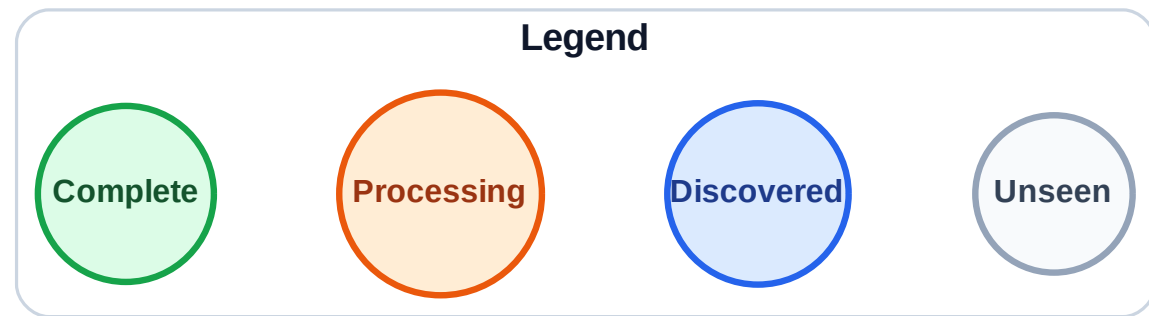
Queue: (empty)

Visited: (none)

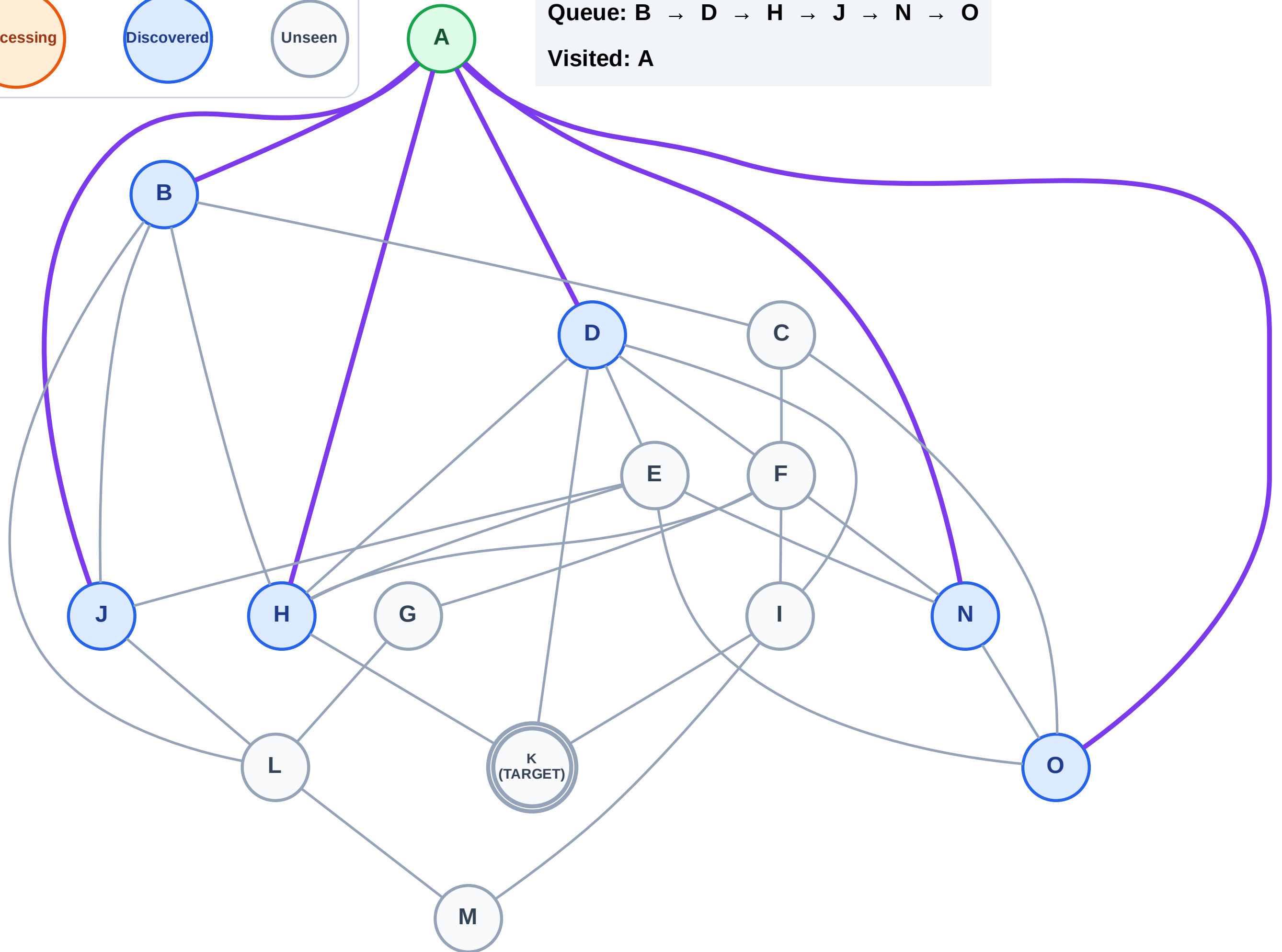


BFS Traversal

Step 3: Discover B, D, H, J, N, O from A

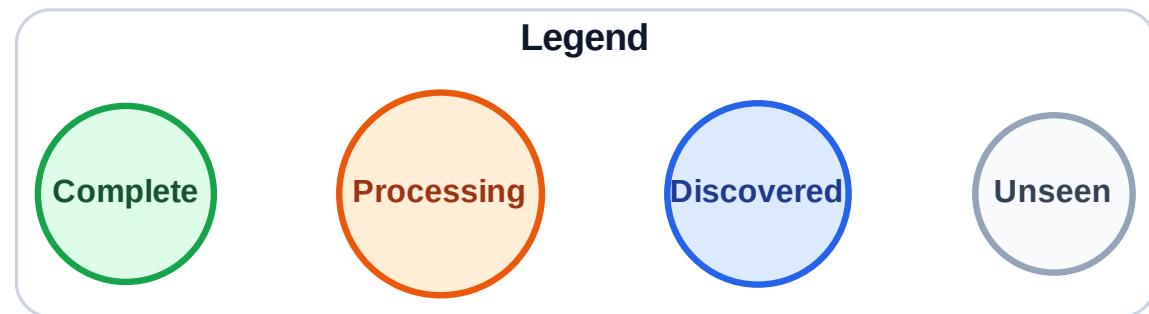


Queue: B → D → H → J → N → O
Visited: A



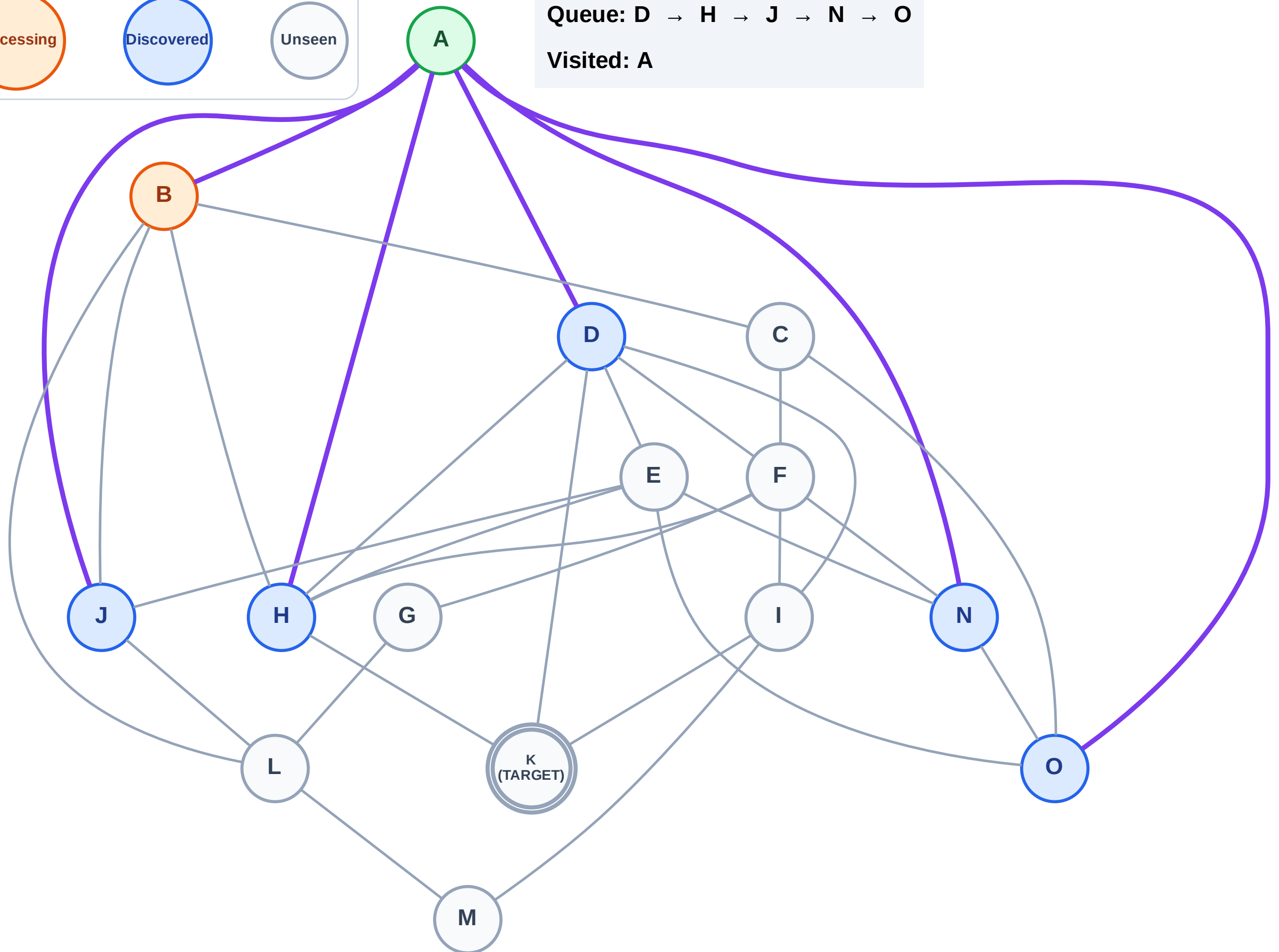
BFS Traversal

Step 4: Process node B



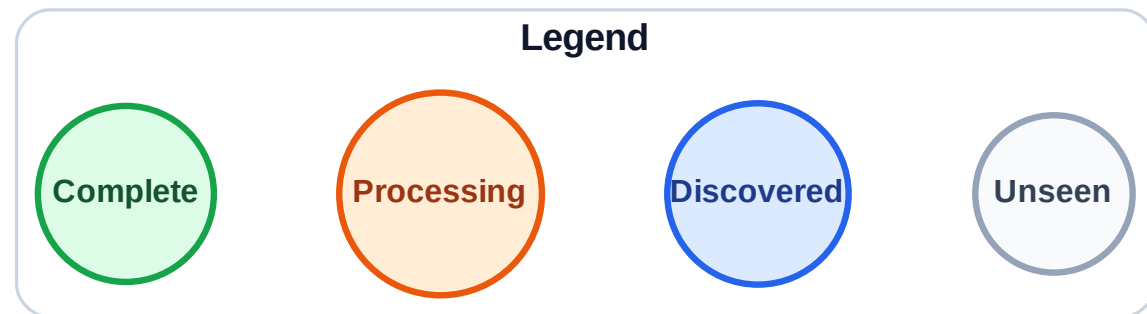
Queue: D → H → J → N → O

Visited: A



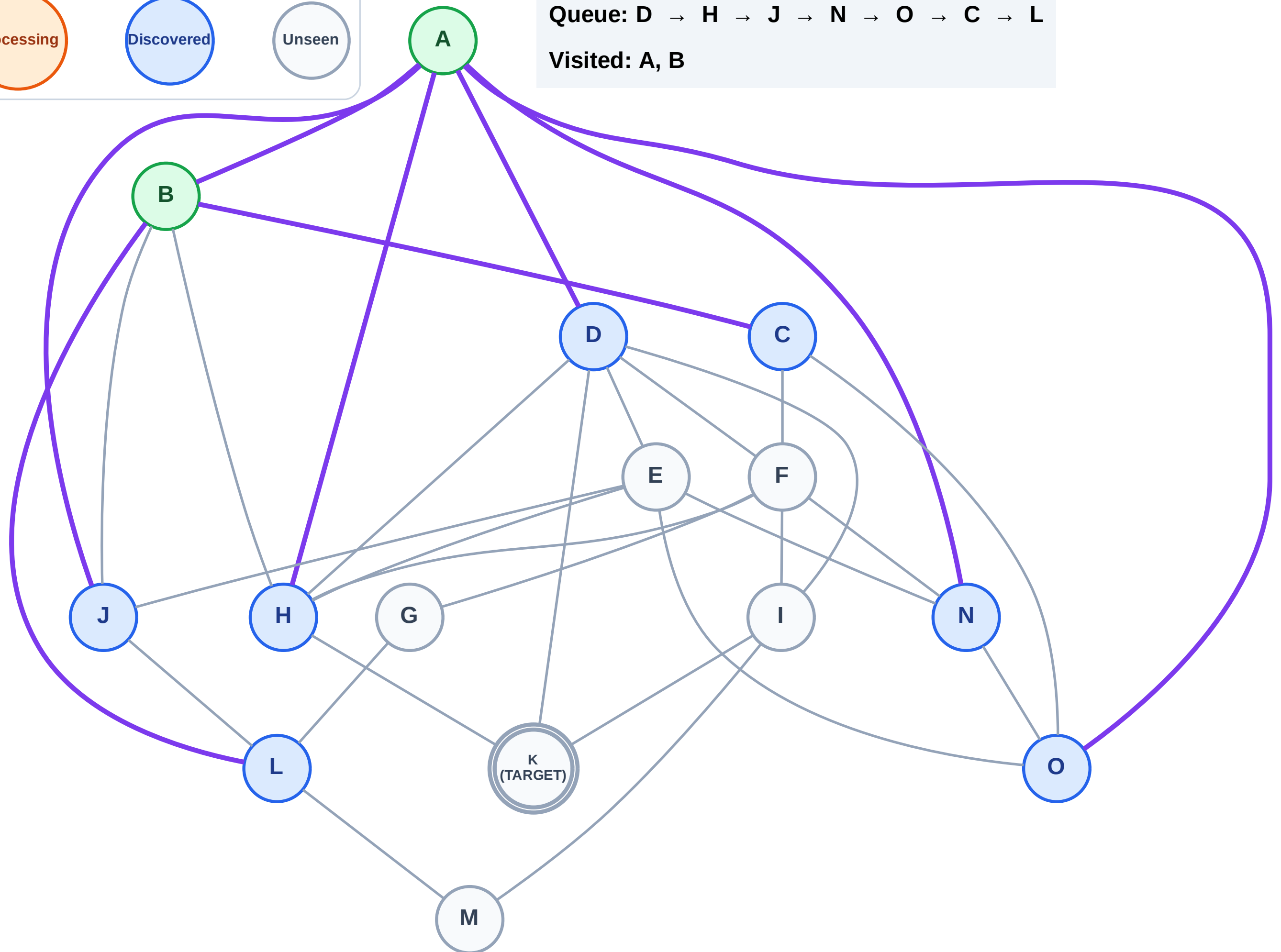
BFS Traversal

Step 5: Discover C, L from B



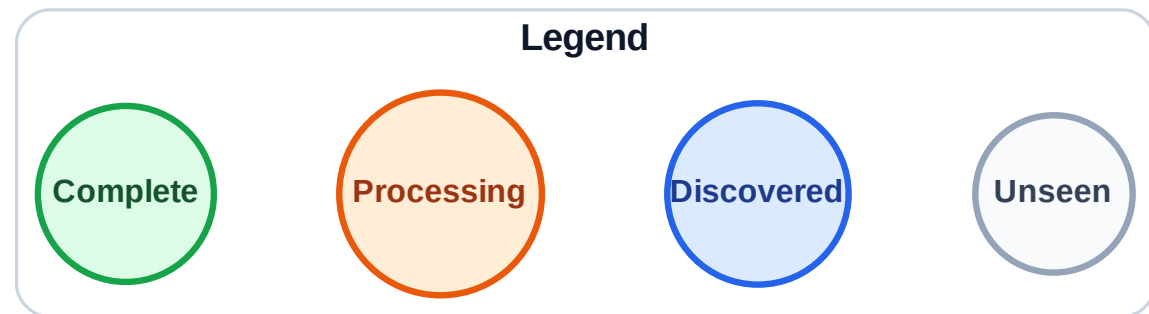
Queue: D → H → J → N → O → C → L

Visited: A, B



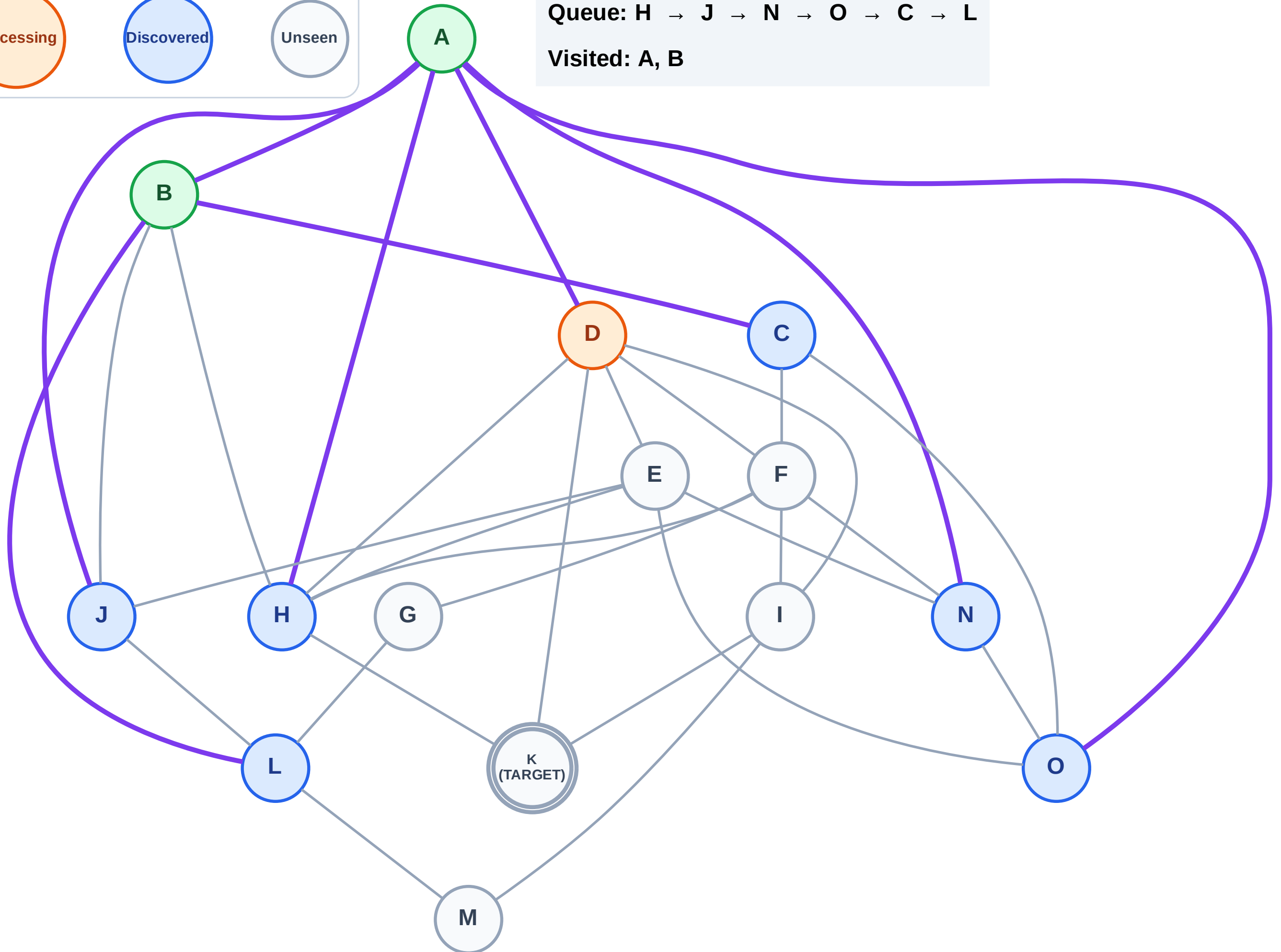
BFS Traversal

Step 6: Process node D



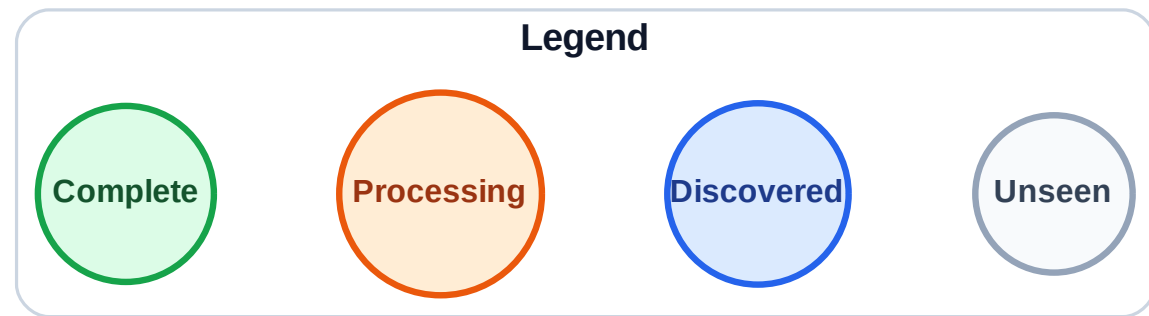
Queue: H → J → N → O → C → L

Visited: A, B



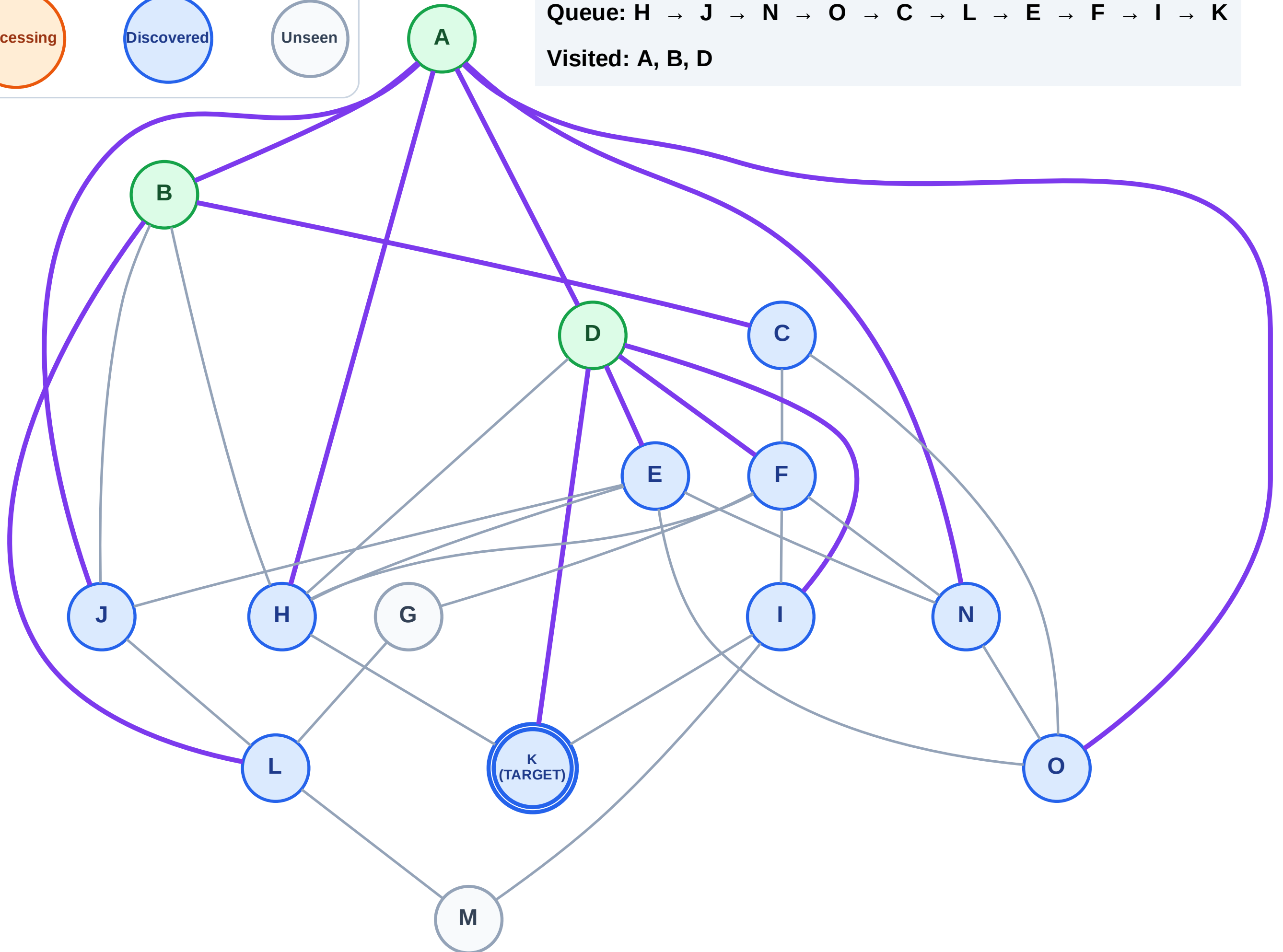
BFS Traversal

Step 7: Discover E, F, I, K from D



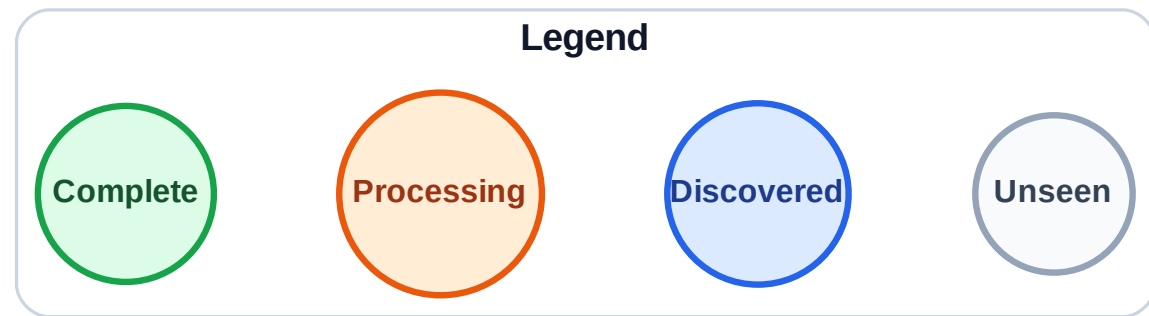
Queue: H → J → N → O → C → L → E → F → I → K

Visited: A, B, D



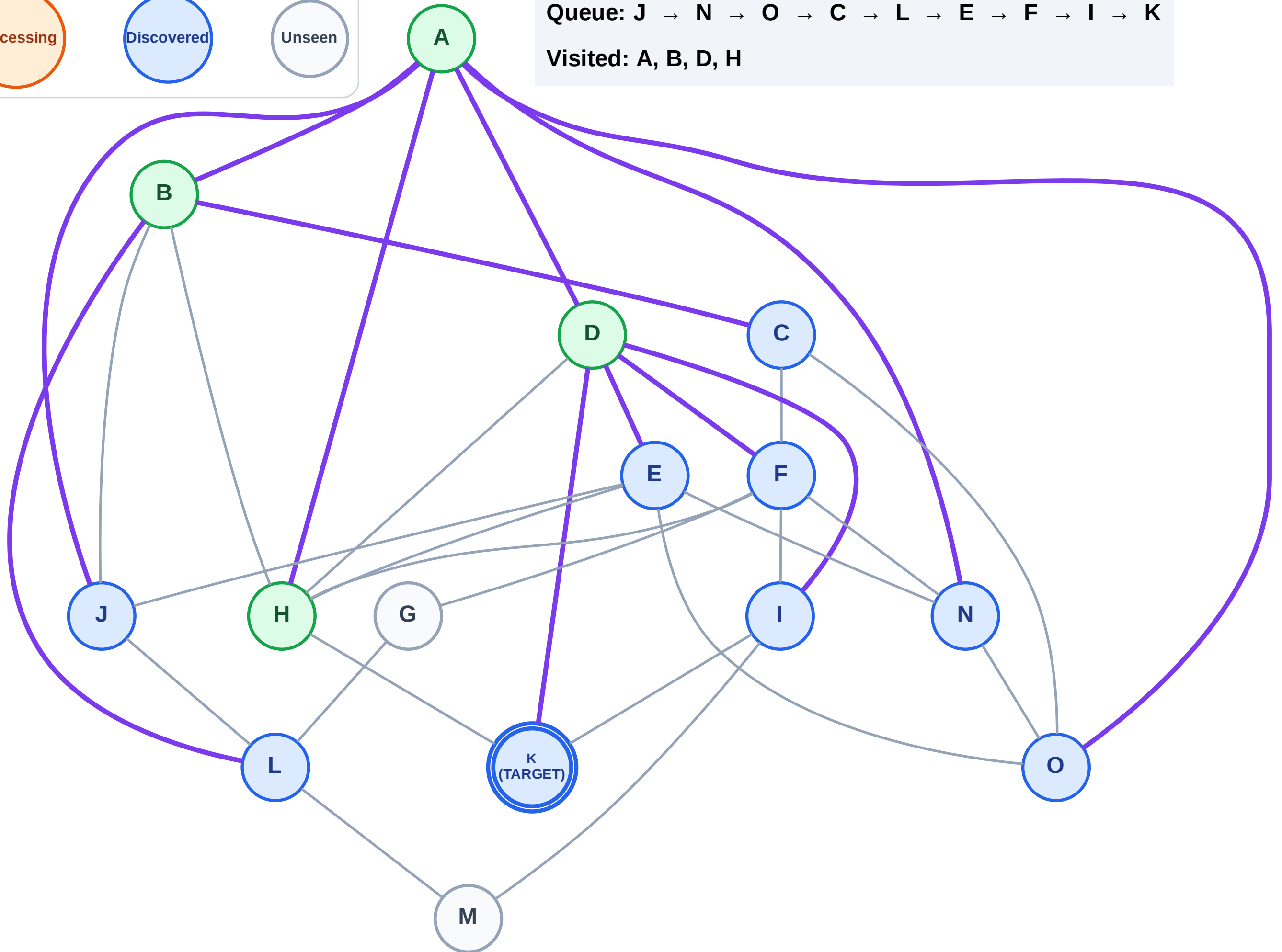
BFS Traversal

Step 9: No new neighbors from H



Queue: J → N → O → C → L → E → F → I → K

Visited: A, B, D, H



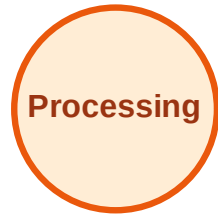
BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



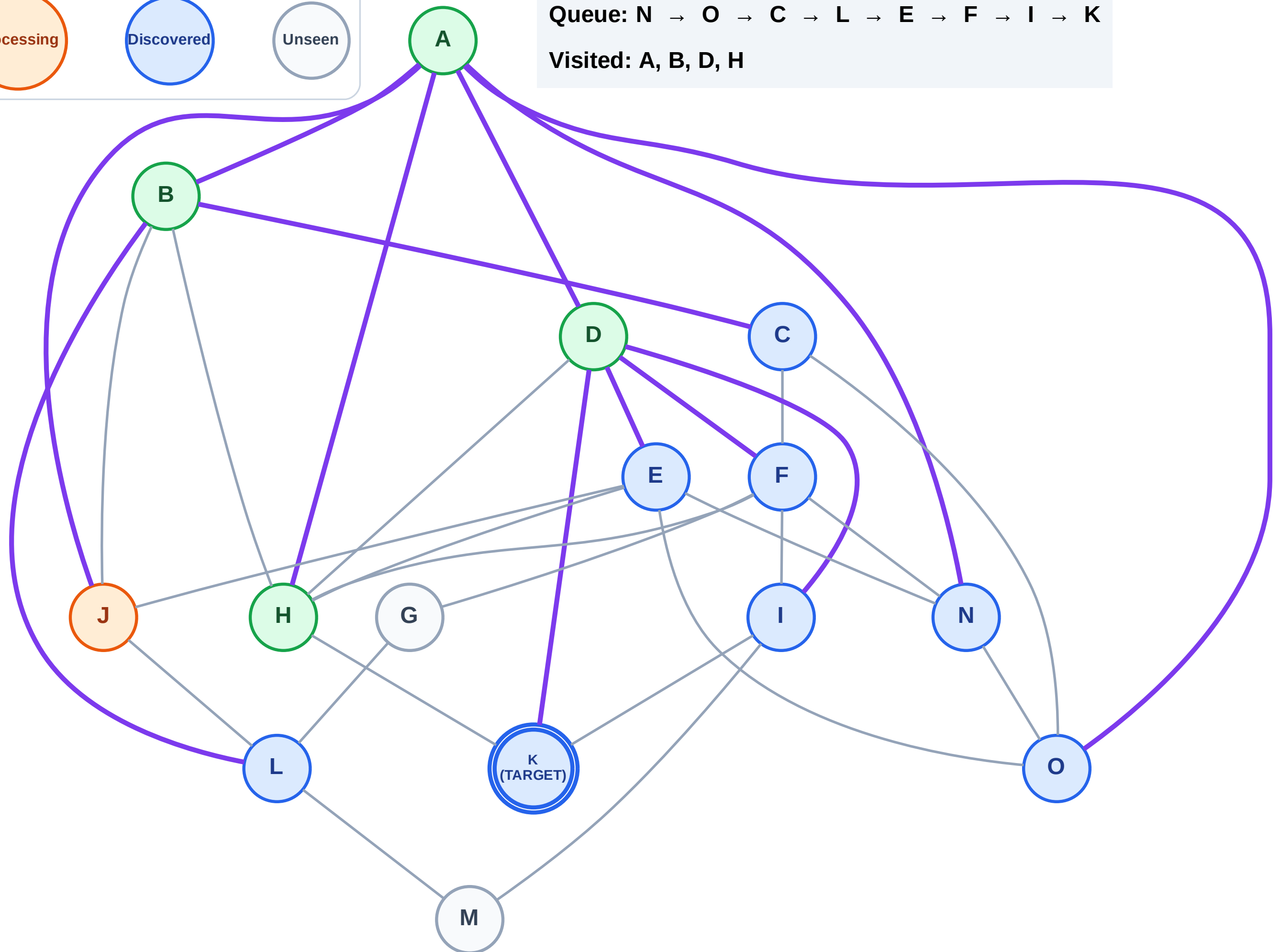
Discovered



Unseen

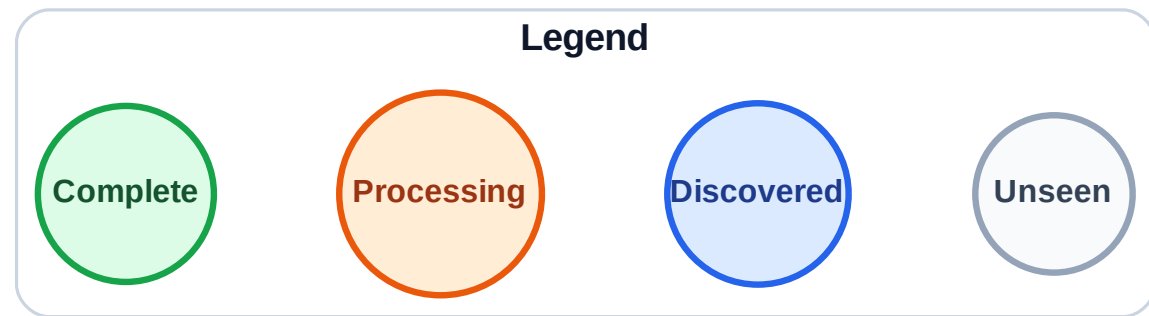
Queue: N → O → C → L → E → F → I → K

Visited: A, B, D, H



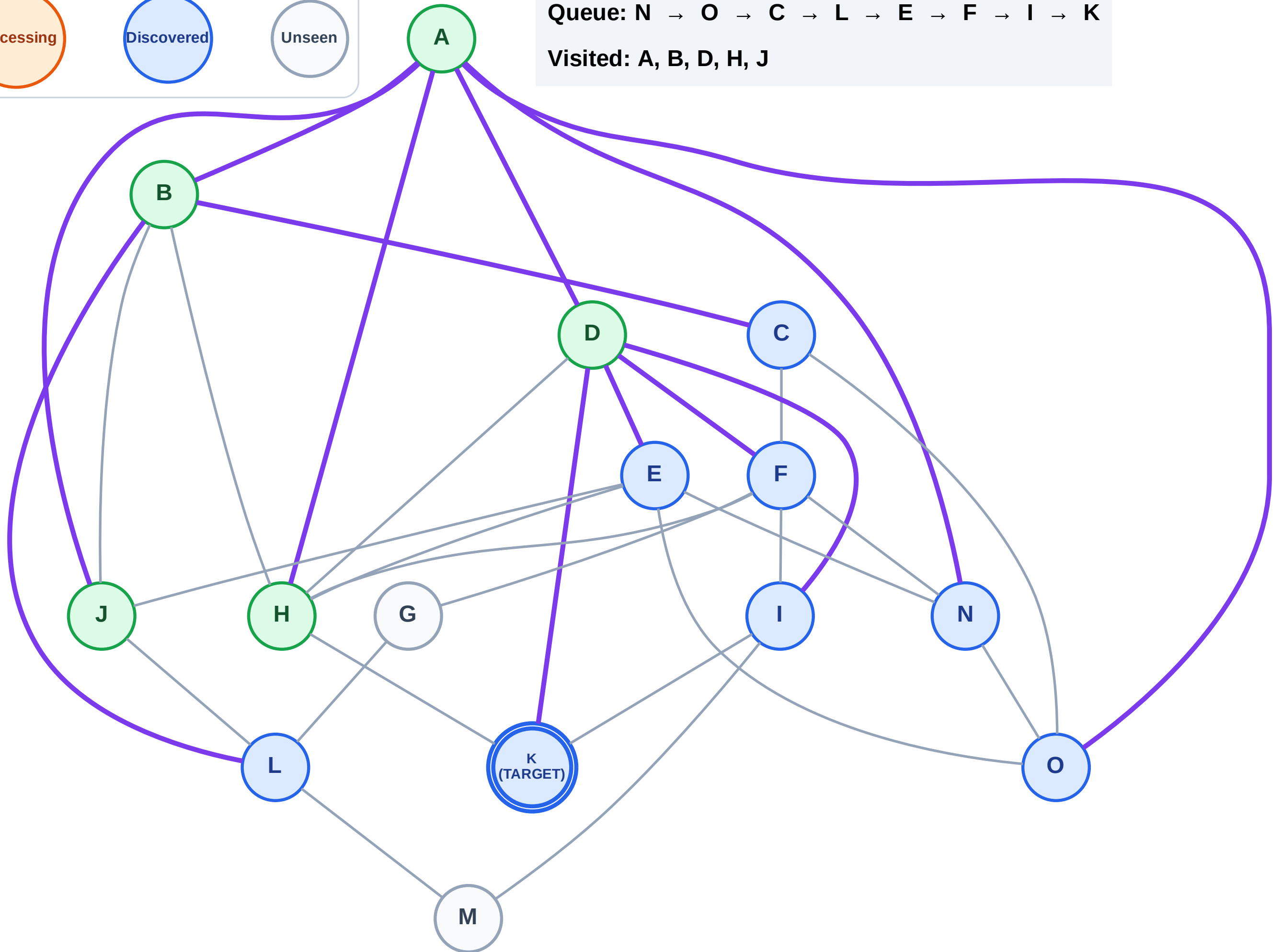
BFS Traversal

Step 11: No new neighbors from J



Queue: N → O → C → L → E → F → I → K

Visited: A, B, D, H, J



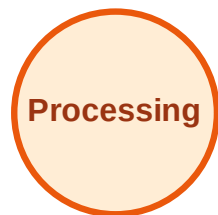
BFS Traversal

Step 12: Process node N

Legend



Complete



Processing



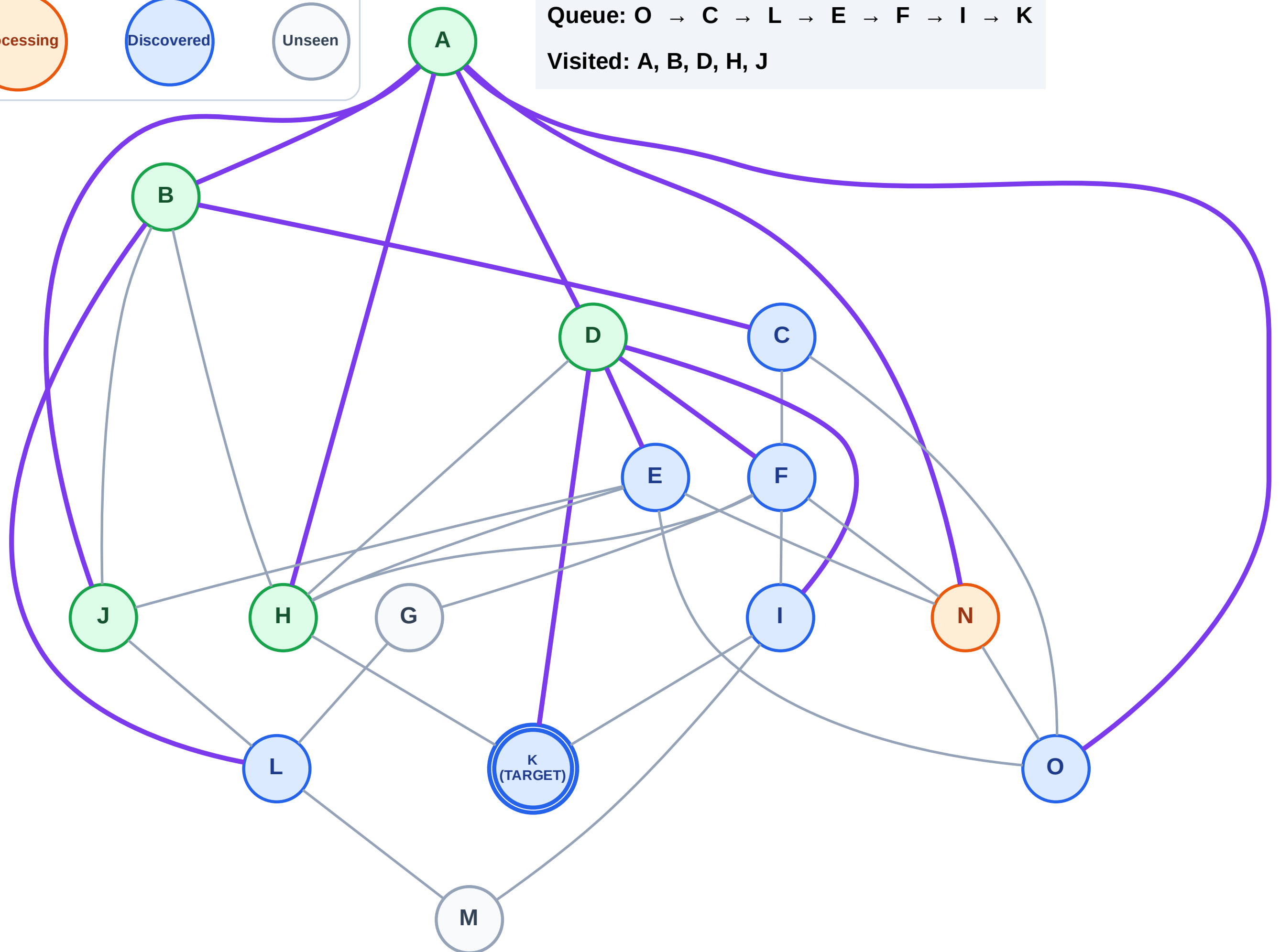
Discovered



Unseen

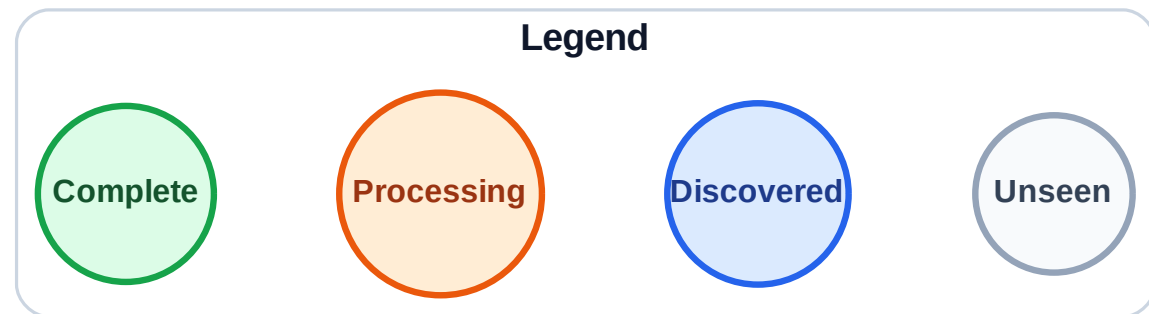
Queue: O → C → L → E → F → I → K

Visited: A, B, D, H, J

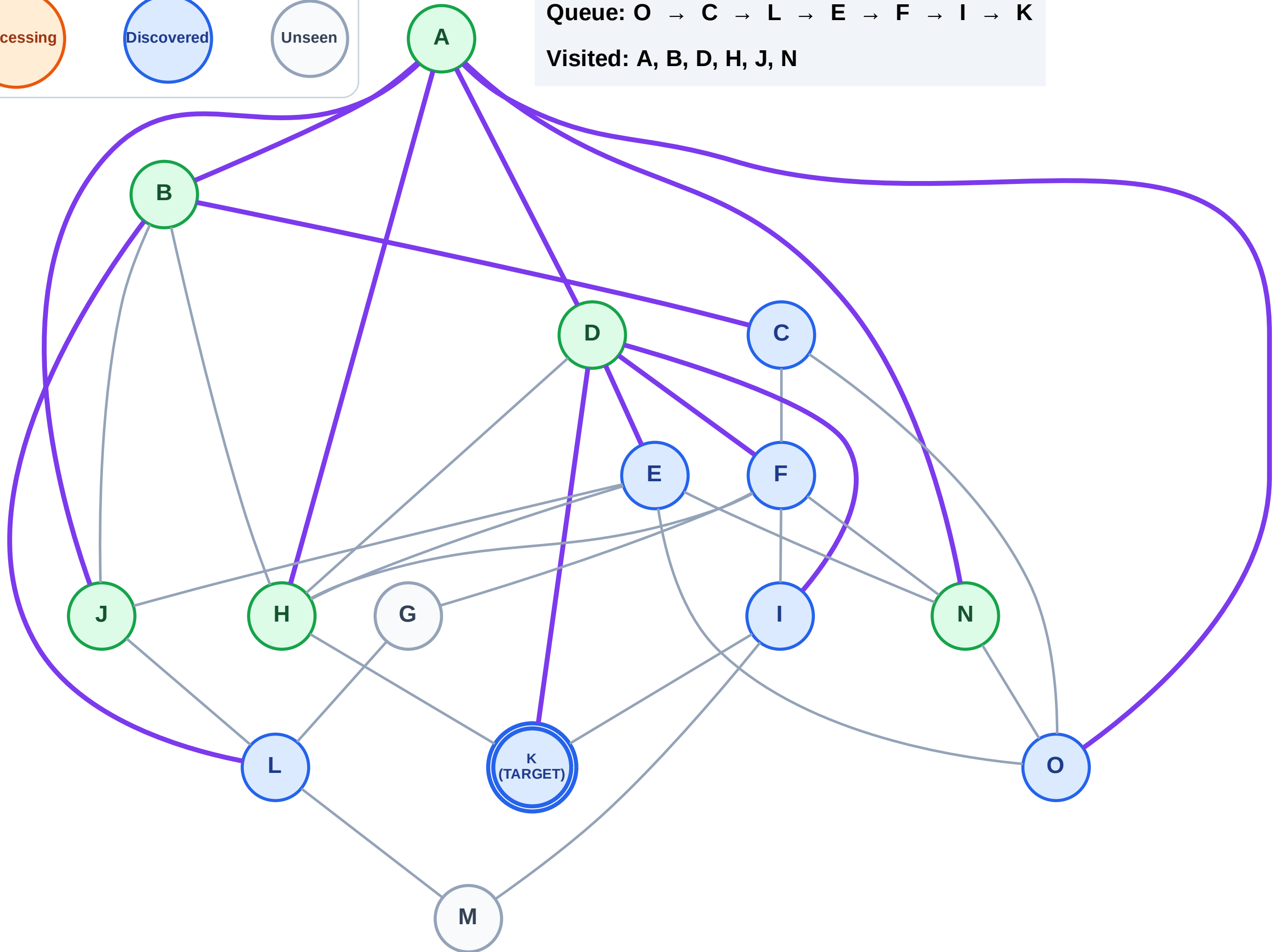


BFS Traversal

Step 13: No new neighbors from N



Queue: O → C → L → E → F → I → K
Visited: A, B, D, H, J, N



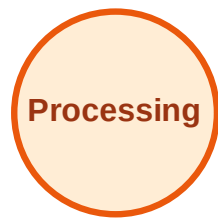
BFS Traversal

Step 14: Process node O

Legend



Complete



Processing



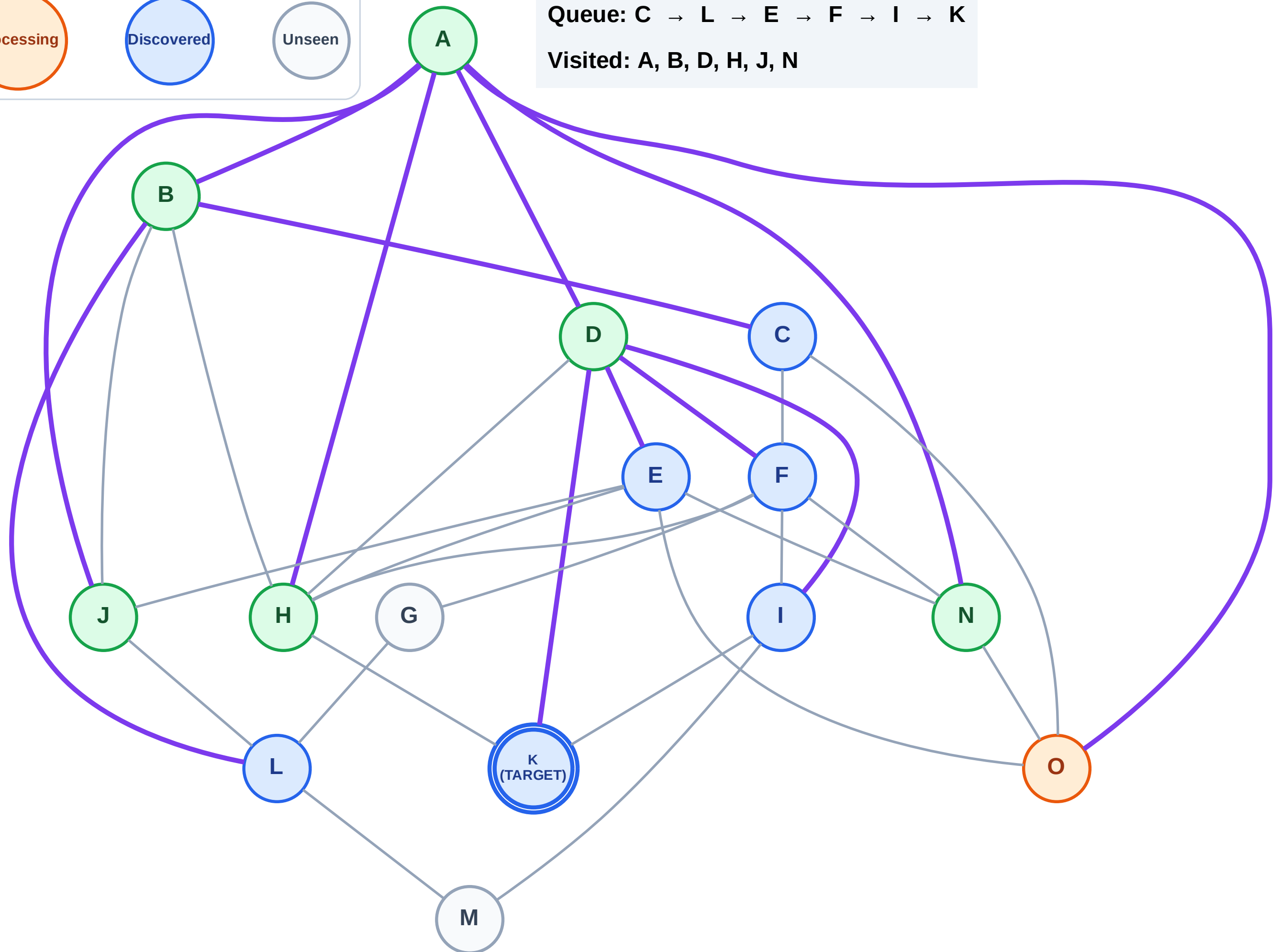
Discovered



Unseen

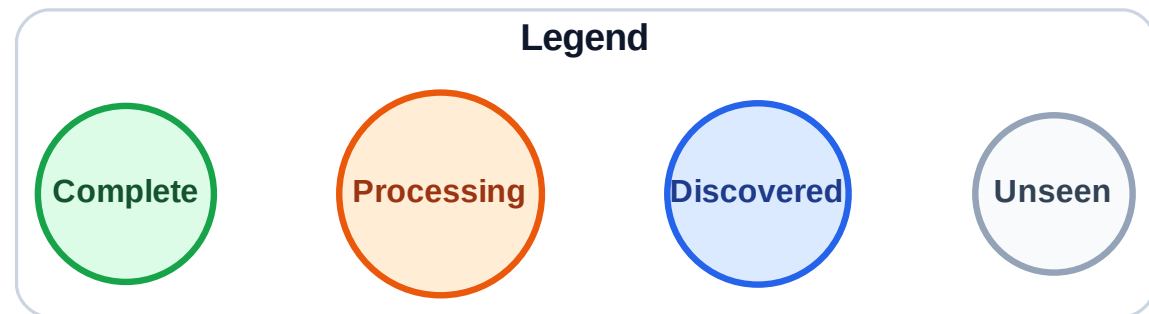
Queue: C → L → E → F → I → K

Visited: A, B, D, H, J, N



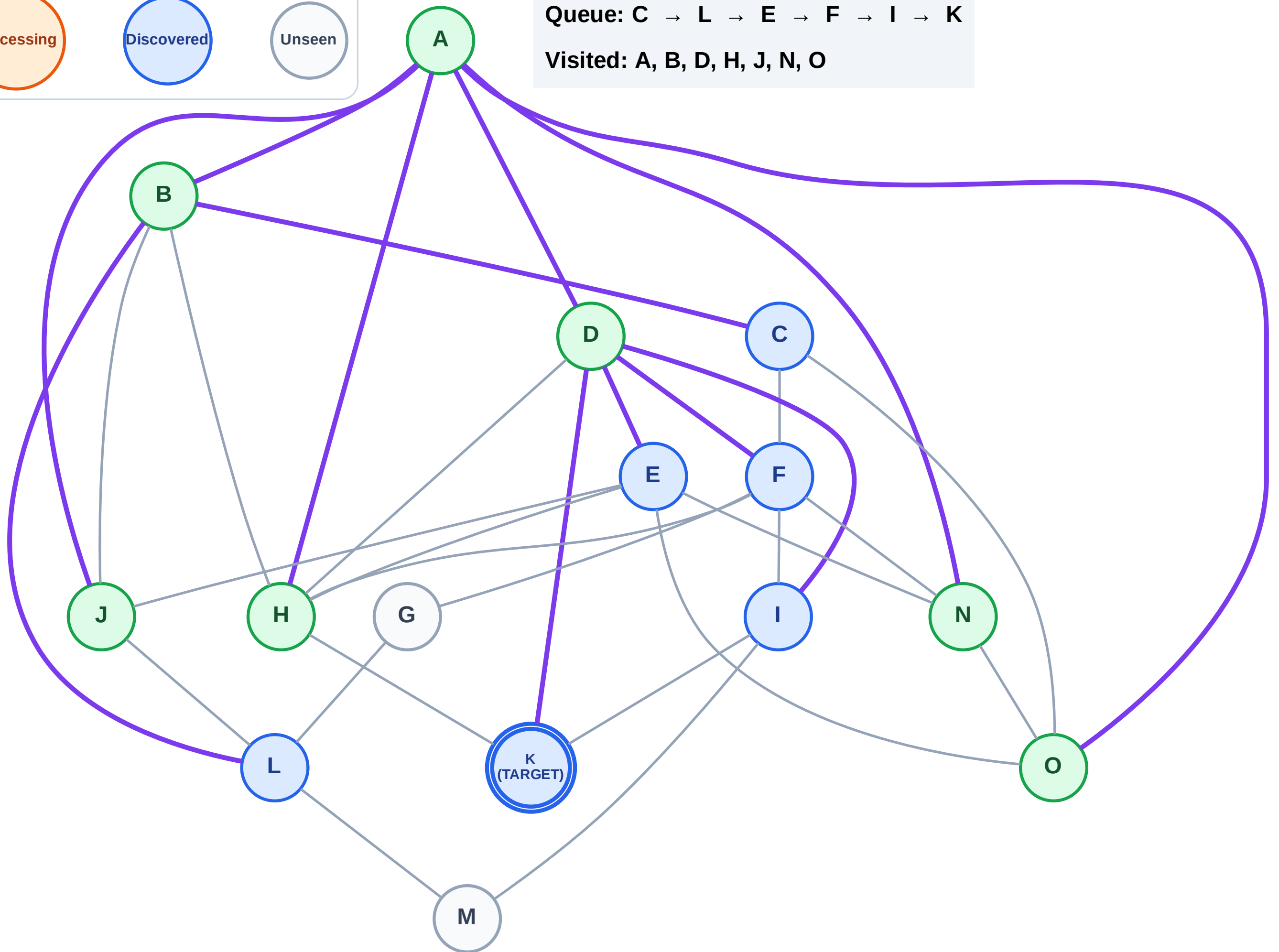
BFS Traversal

Step 15: No new neighbors from O



Queue: C → L → E → F → I → K

Visited: A, B, D, H, J, N, O



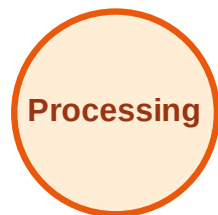
BFS Traversal

Step 16: Process node C

Legend



Complete



Processing



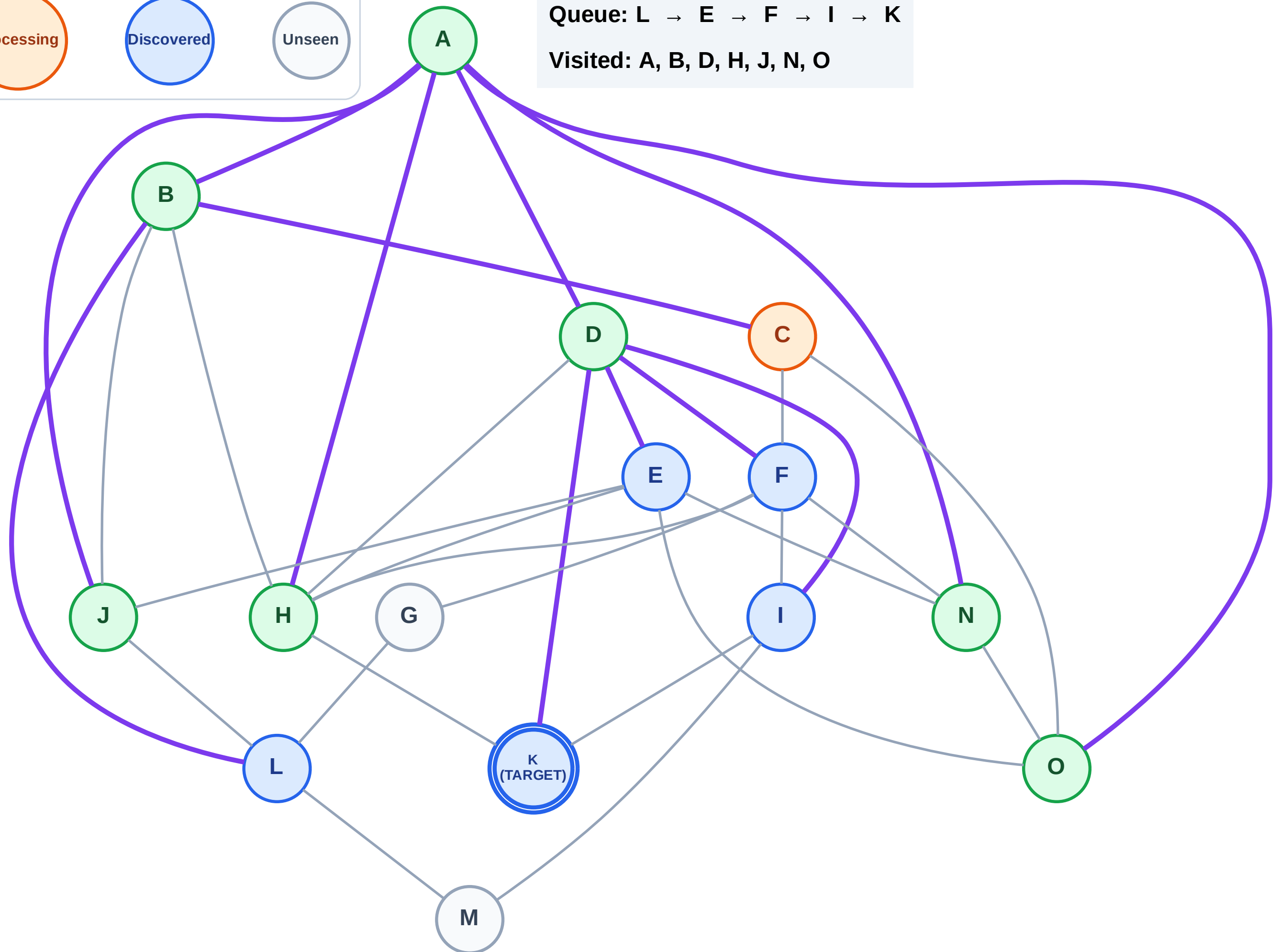
Discovered



Unseen

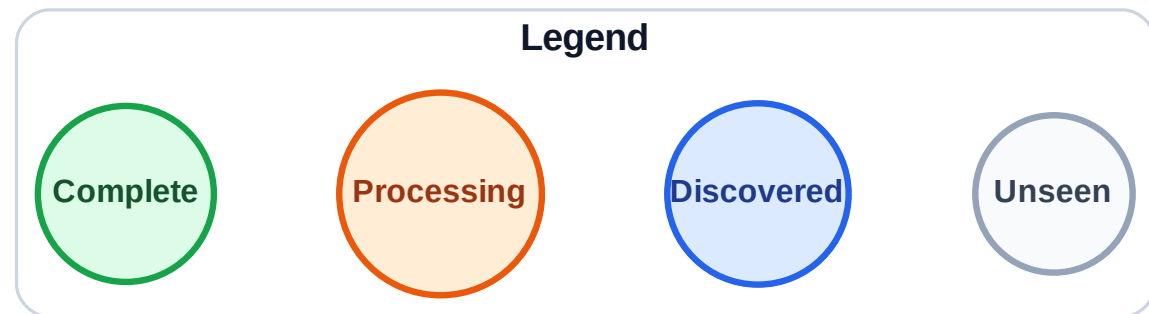
Queue: L → E → F → I → K

Visited: A, B, D, H, J, N, O



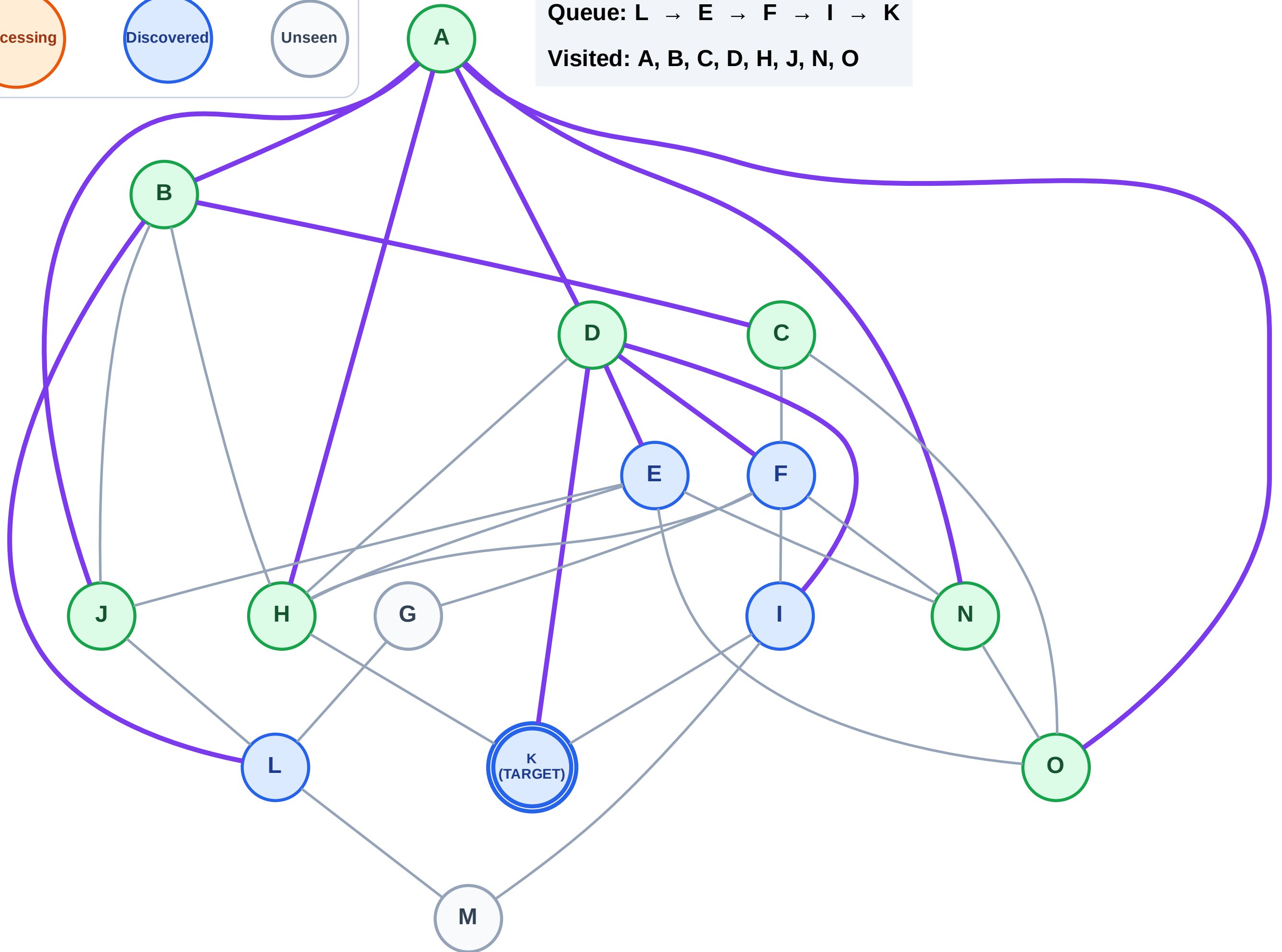
BFS Traversal

Step 17: No new neighbors from C



Queue: L → E → F → I → K

Visited: A, B, C, D, H, J, N, O



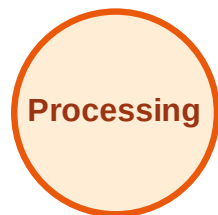
BFS Traversal

Step 18: Process node L

Legend



Complete



Processing



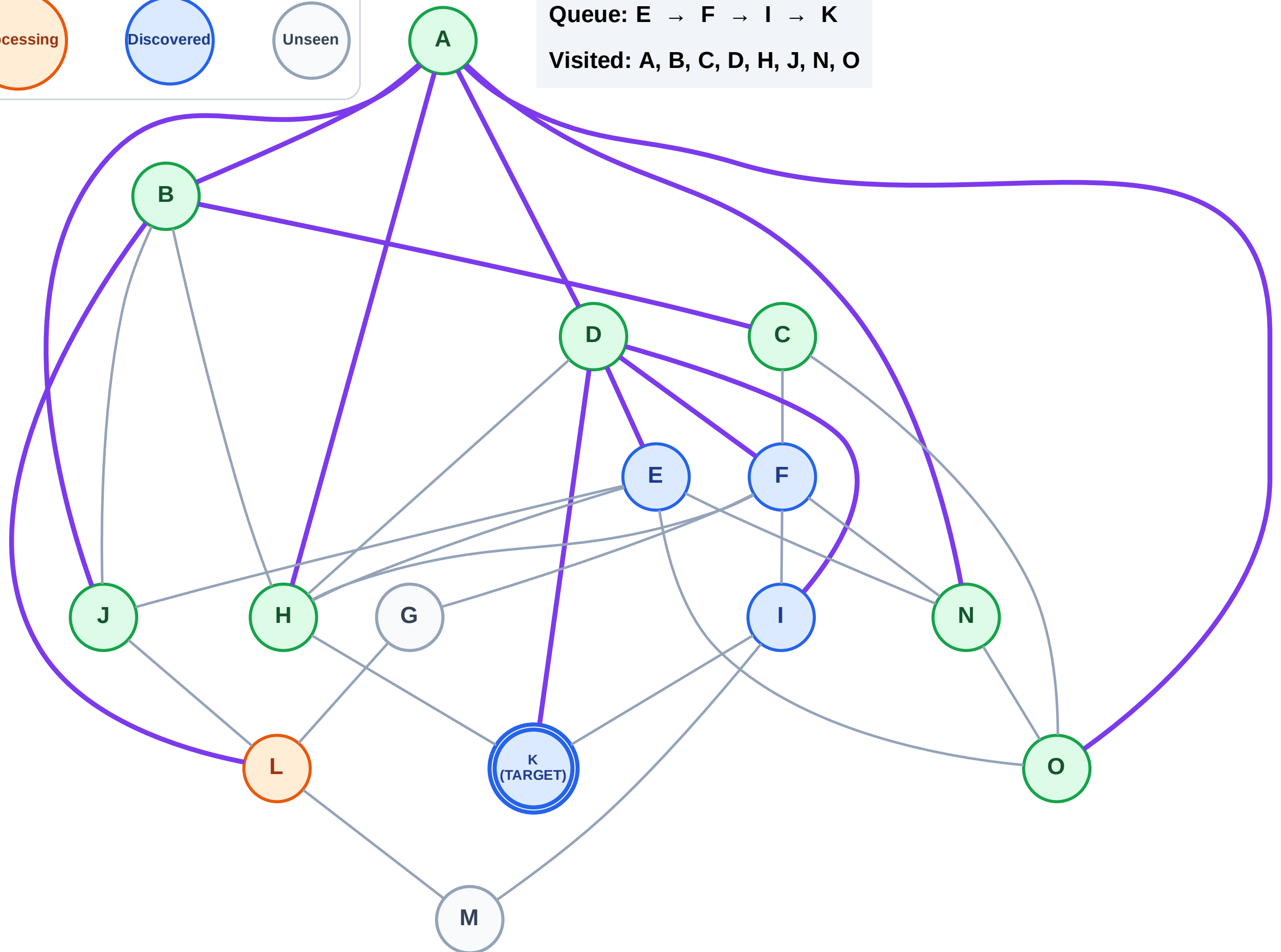
Discovered



Unseen

Queue: E → F → I → K

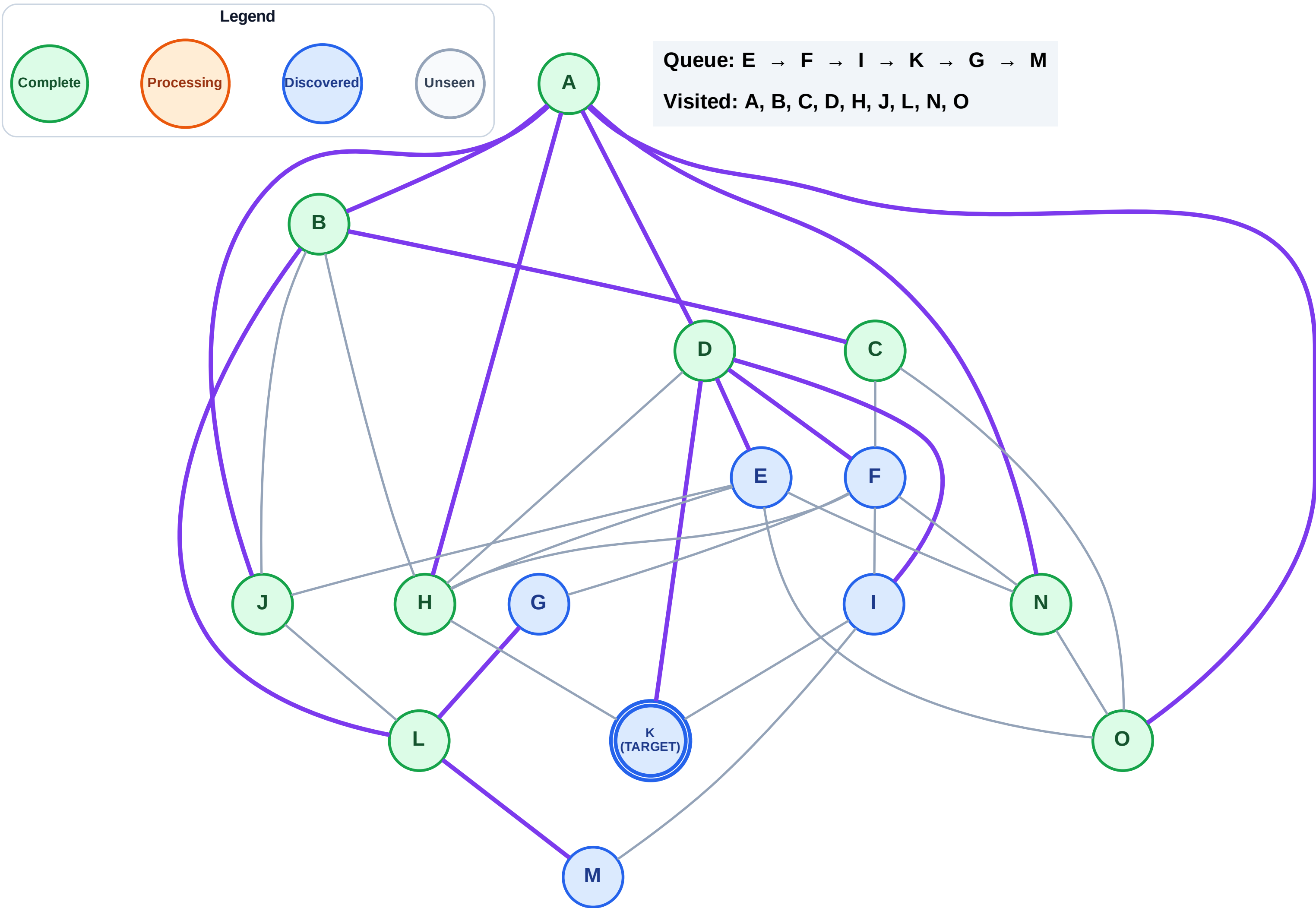
Visited: A, B, C, D, H, J, N, O



Step 19: Discover G , M from L

Queue: E → F → I → K → G → M

Visited: A, B, C, D, H, J, L, N, O



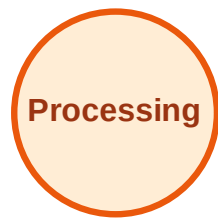
BFS Traversal

Step 20: Process node E

Legend



Complete



Processing



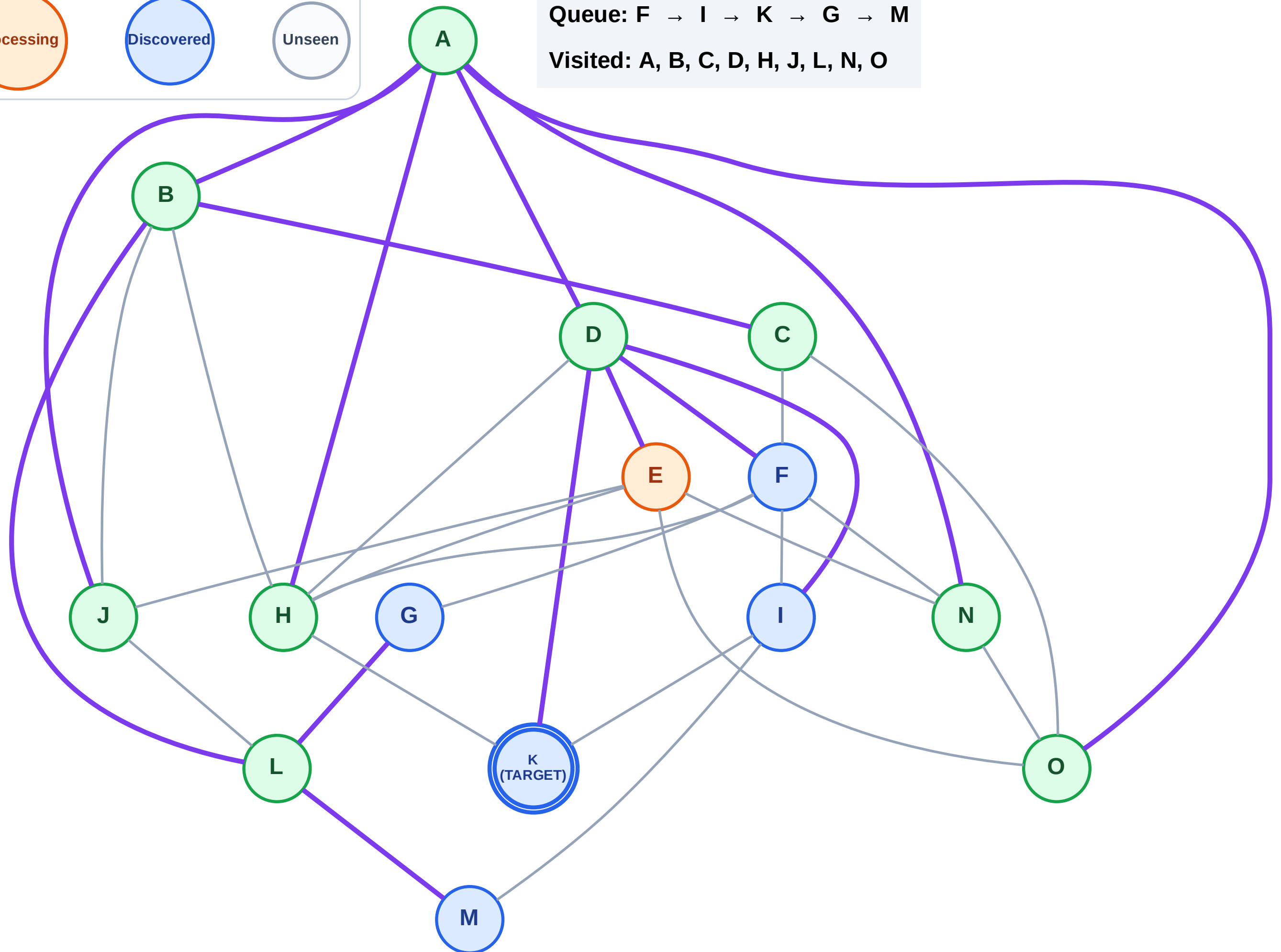
Discovered



Unseen

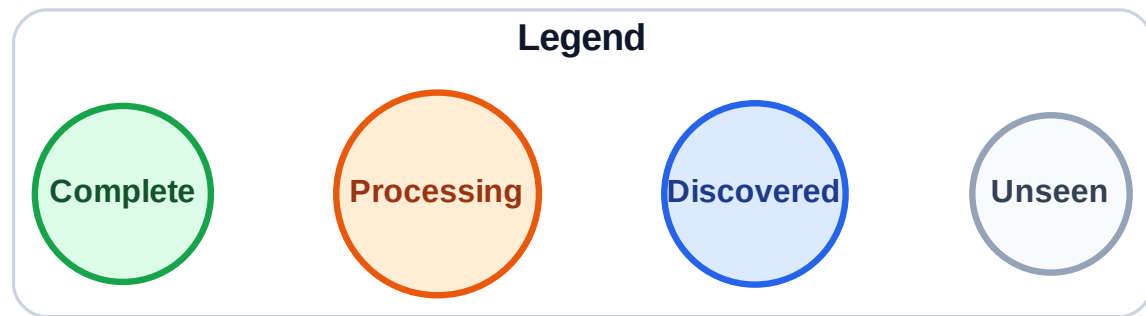
Queue: F → I → K → G → M

Visited: A, B, C, D, H, J, L, N, O



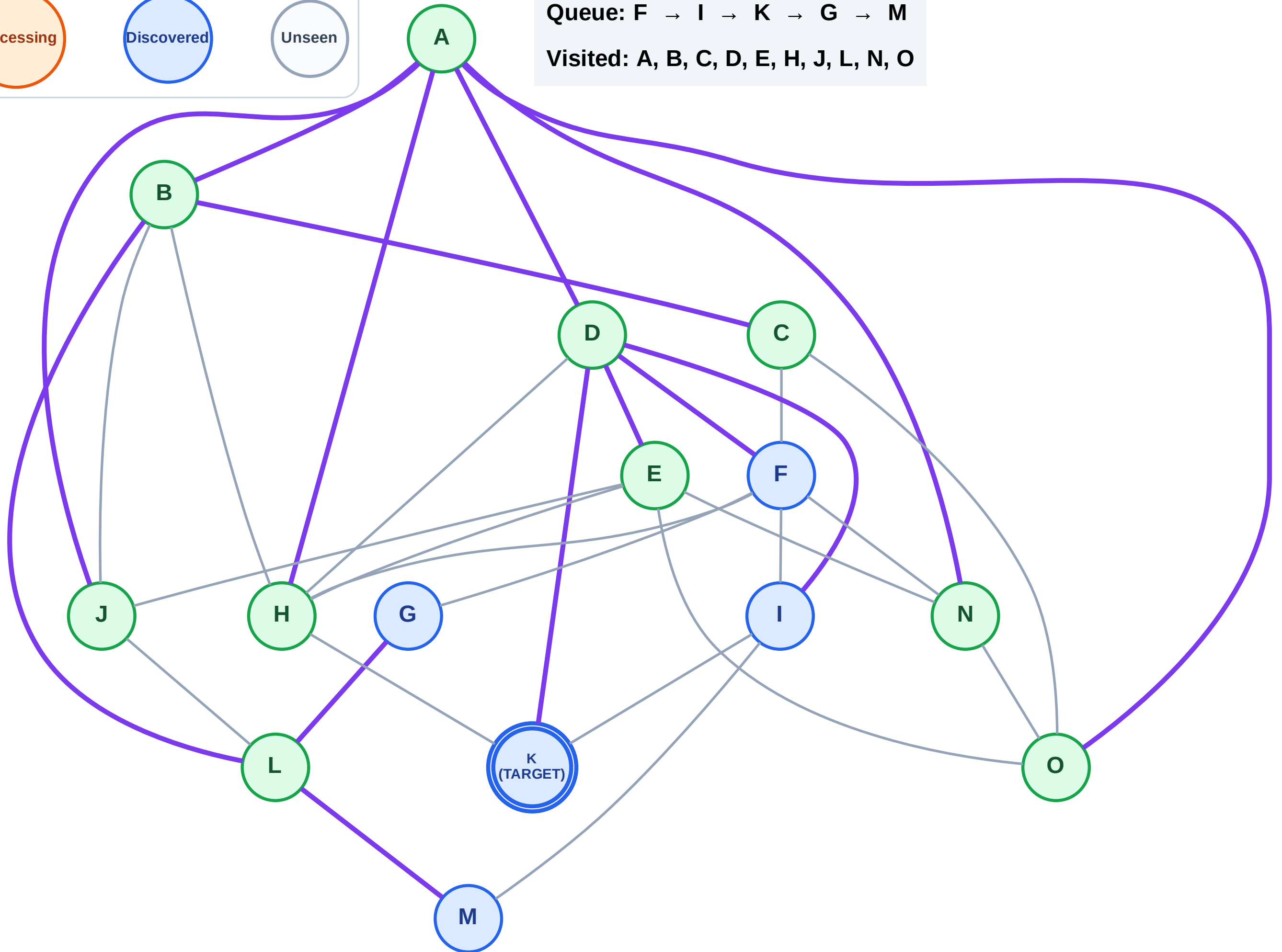
BFS Traversal

Step 21: No new neighbors from E



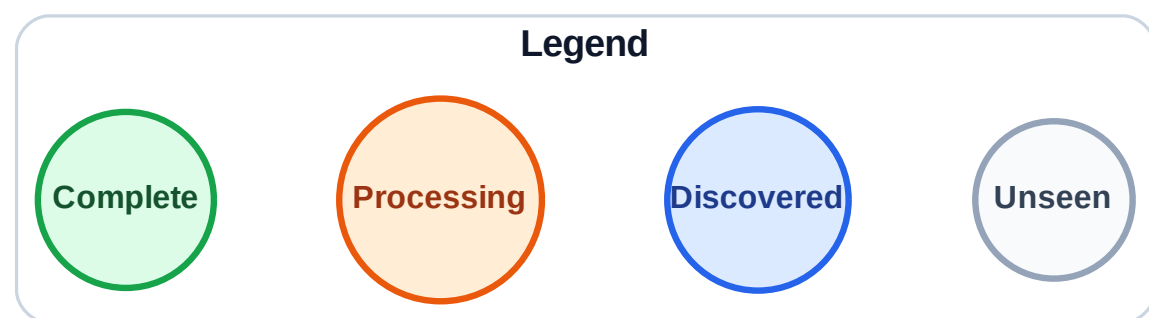
Queue: F → I → K → G → M

Visited: A, B, C, D, E, H, J, L, N, O



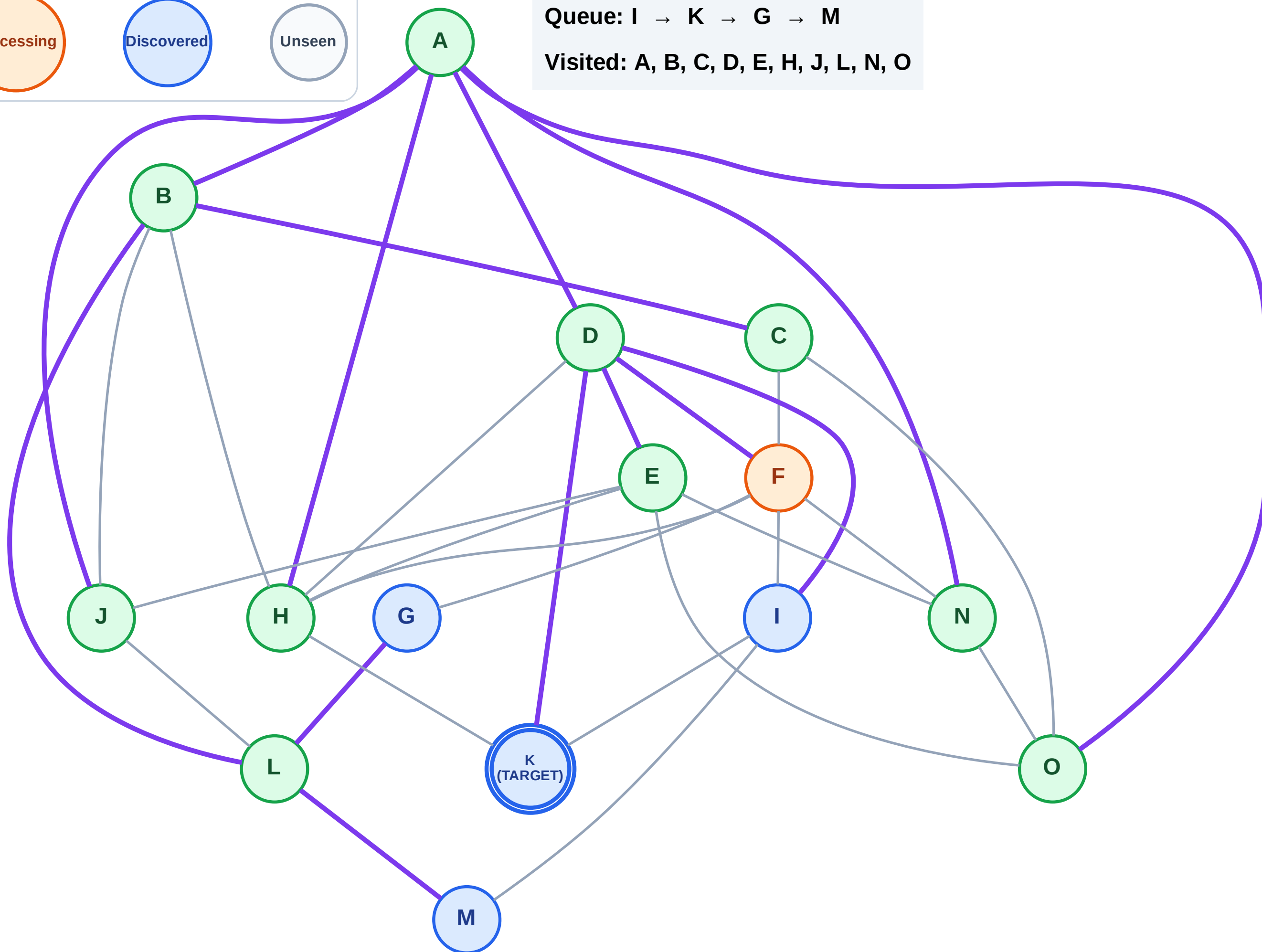
BFS Traversal

Step 22: Process node F



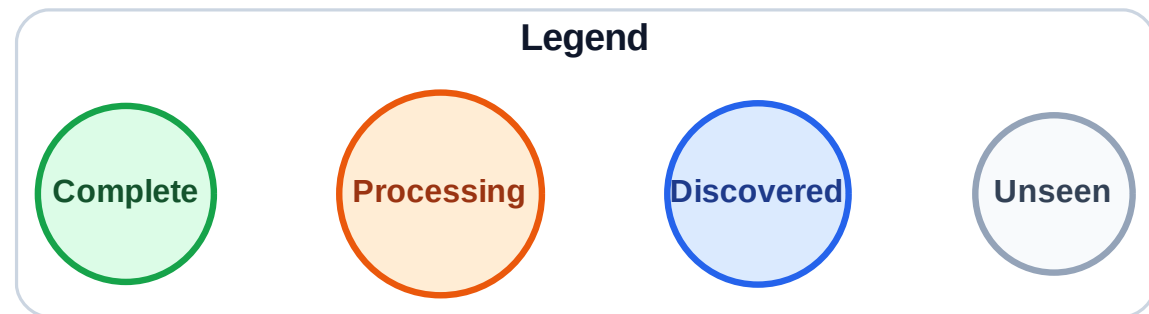
Queue: I → K → G → M

Visited: A, B, C, D, E, H, J, L, N, O



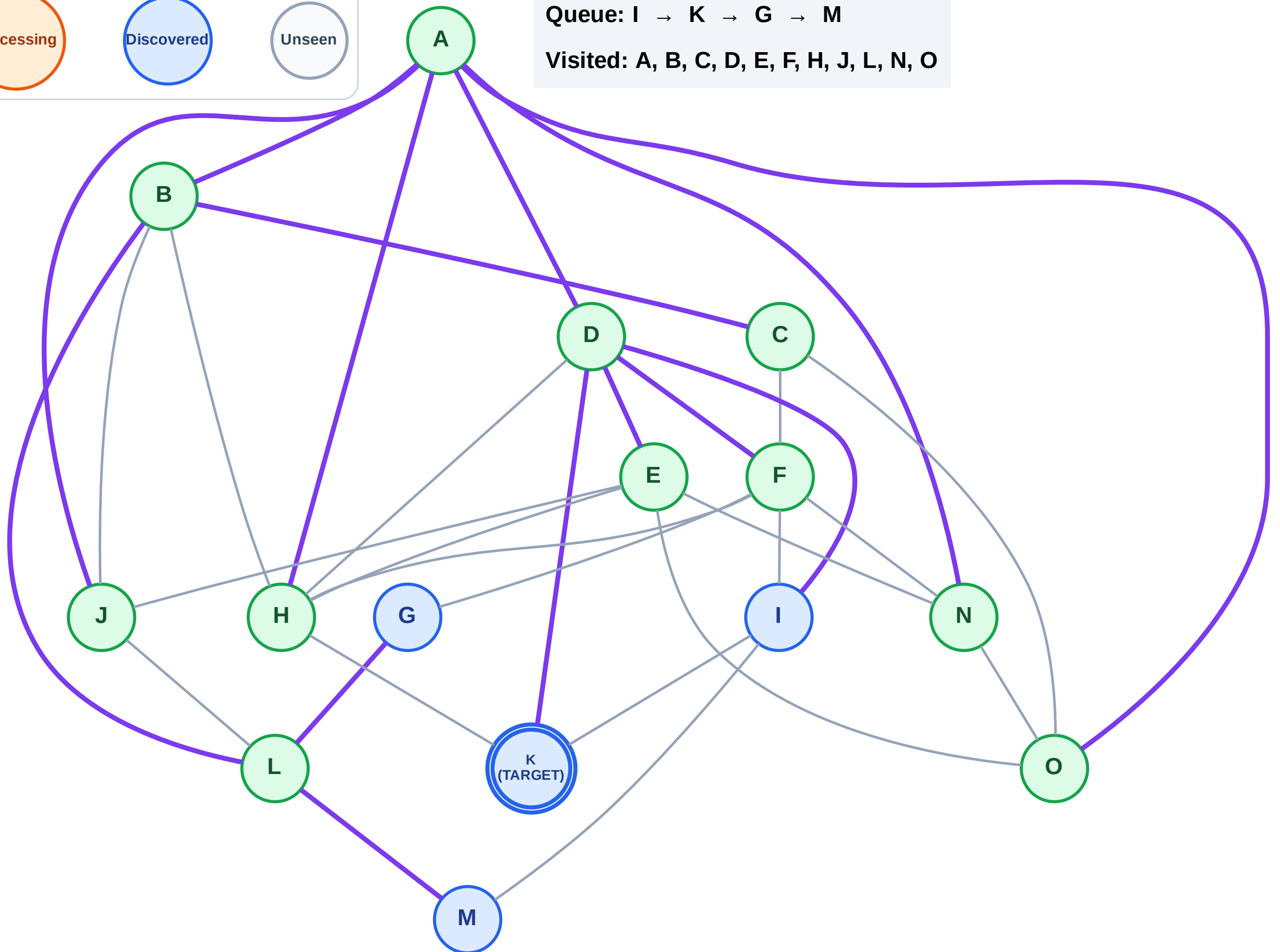
BFS Traversal

Step 23: No new neighbors from F



Queue: I → K → G → M

Visited: A, B, C, D, E, F, H, J, L, N, O



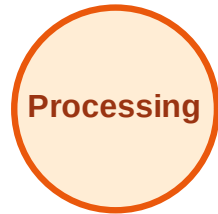
BFS Traversal

Step 24: Process node I

Legend



Complete



Processing



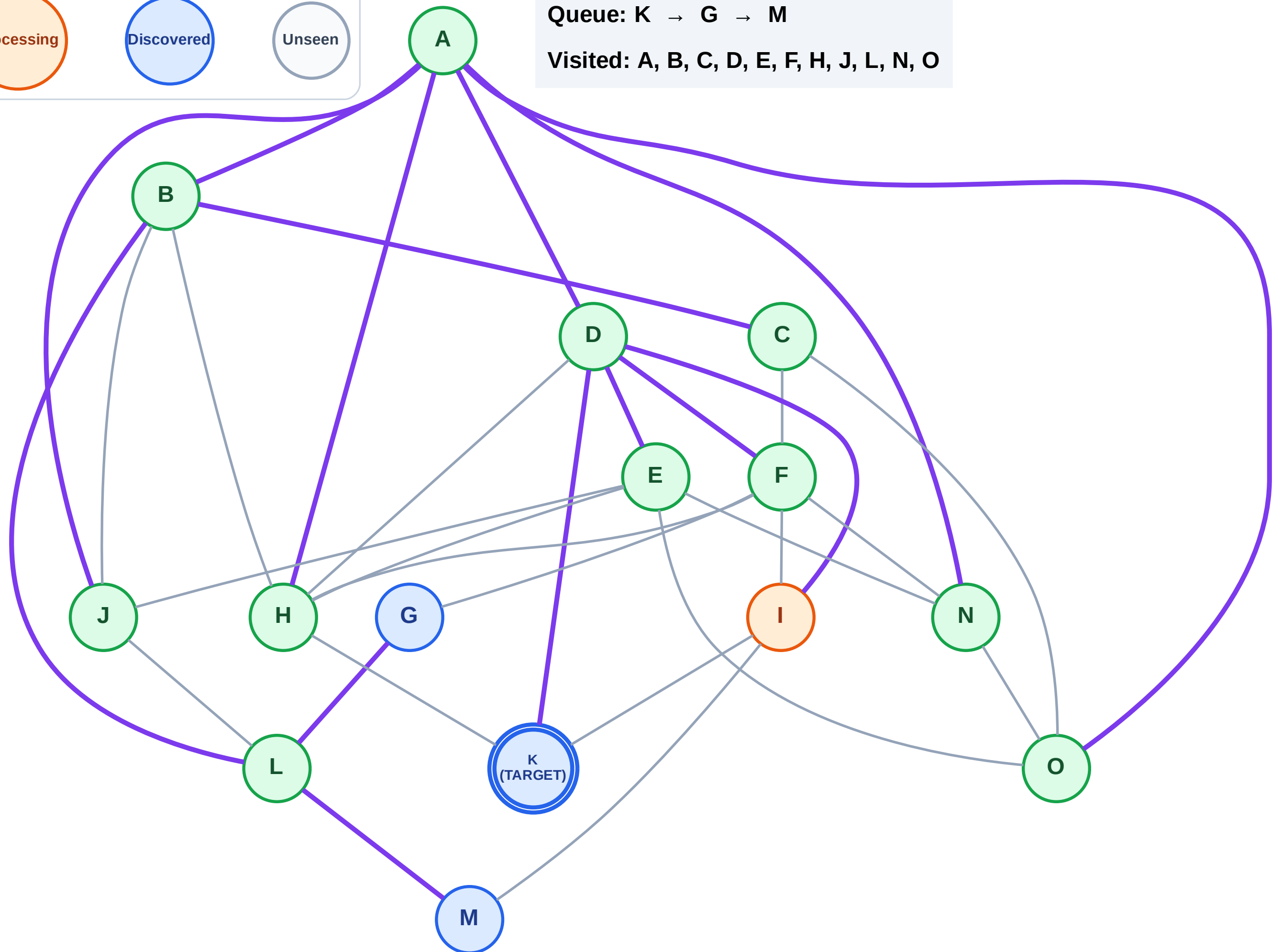
Discovered



Unseen

Queue: K → G → M

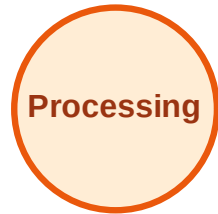
Visited: A, B, C, D, E, F, H, J, L, N, O



BFS Traversal

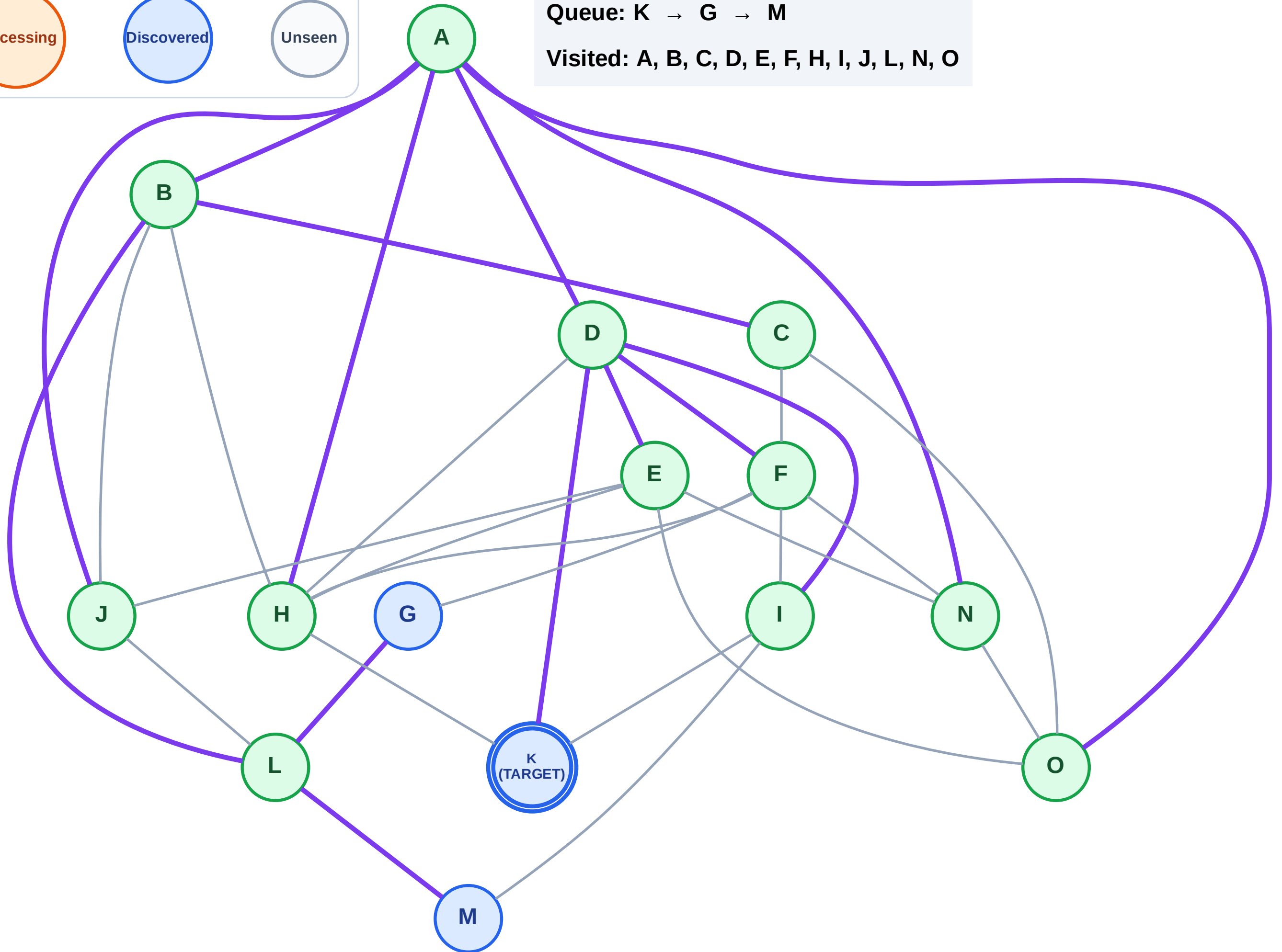
Step 25: No new neighbors from I

Legend



Queue: K → G → M

Visited: A, B, C, D, E, F, H, I, J, L, N, O



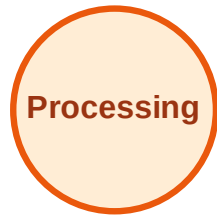
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



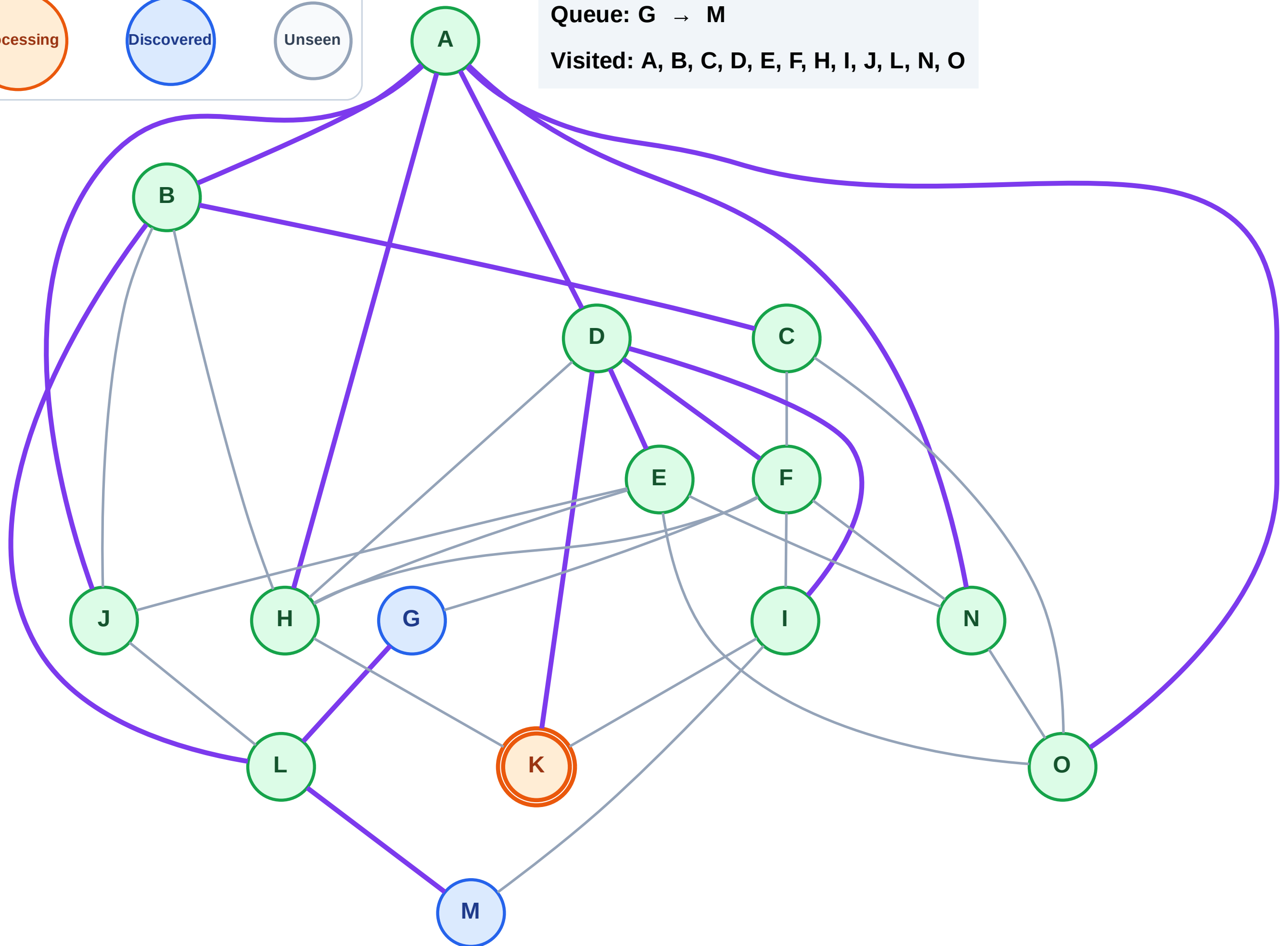
Discovered



Unseen

Queue: G → M

Visited: A, B, C, D, E, F, H, I, J, L, N, O



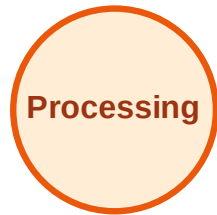
BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Complete



Processing



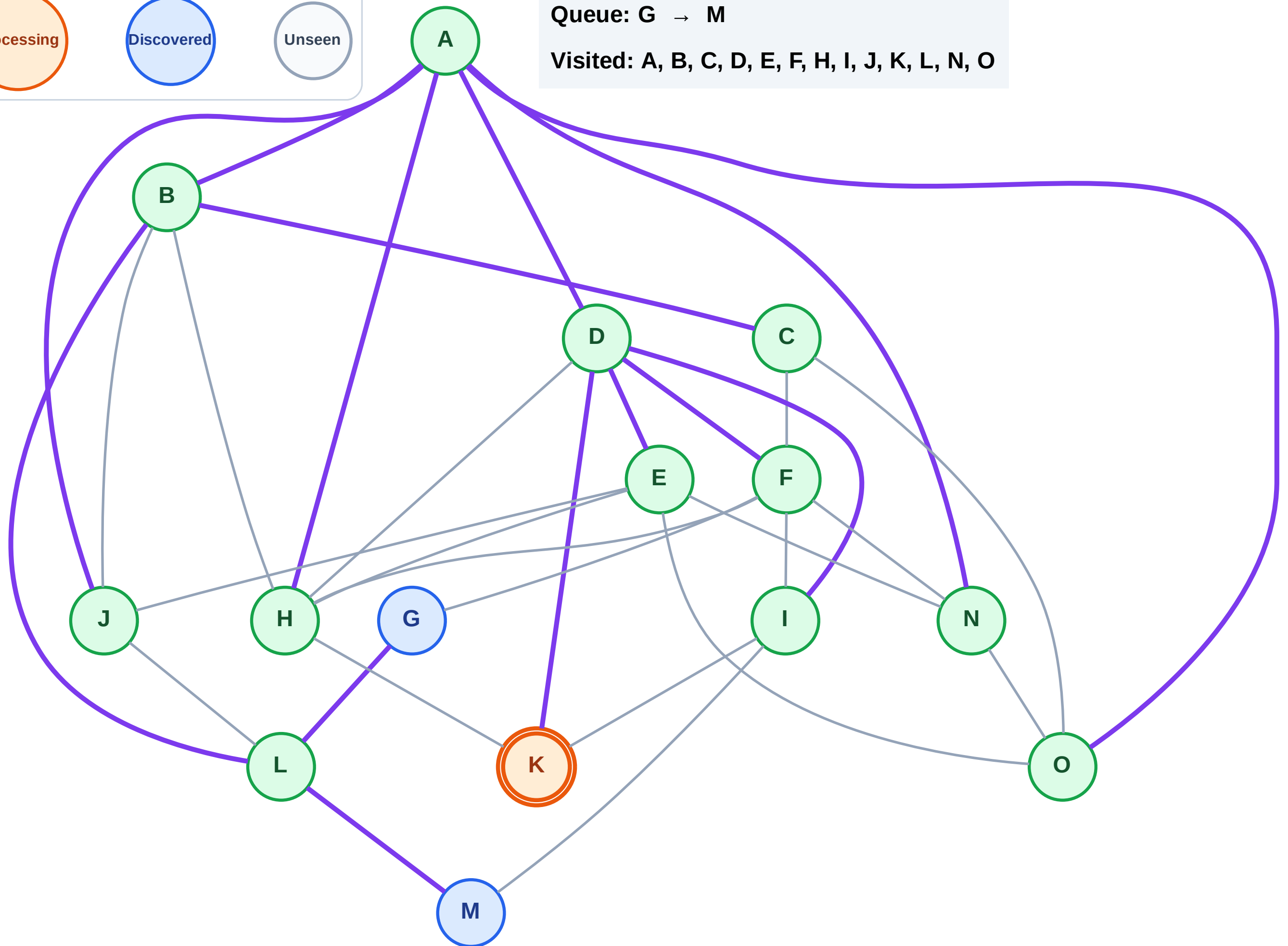
Discovered



Unseen

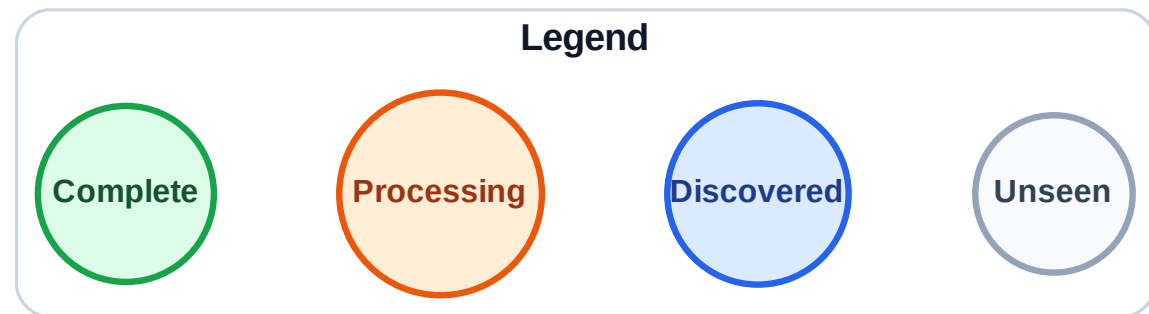
Queue: G → M

Visited: A, B, C, D, E, F, H, I, J, K, L, N, O

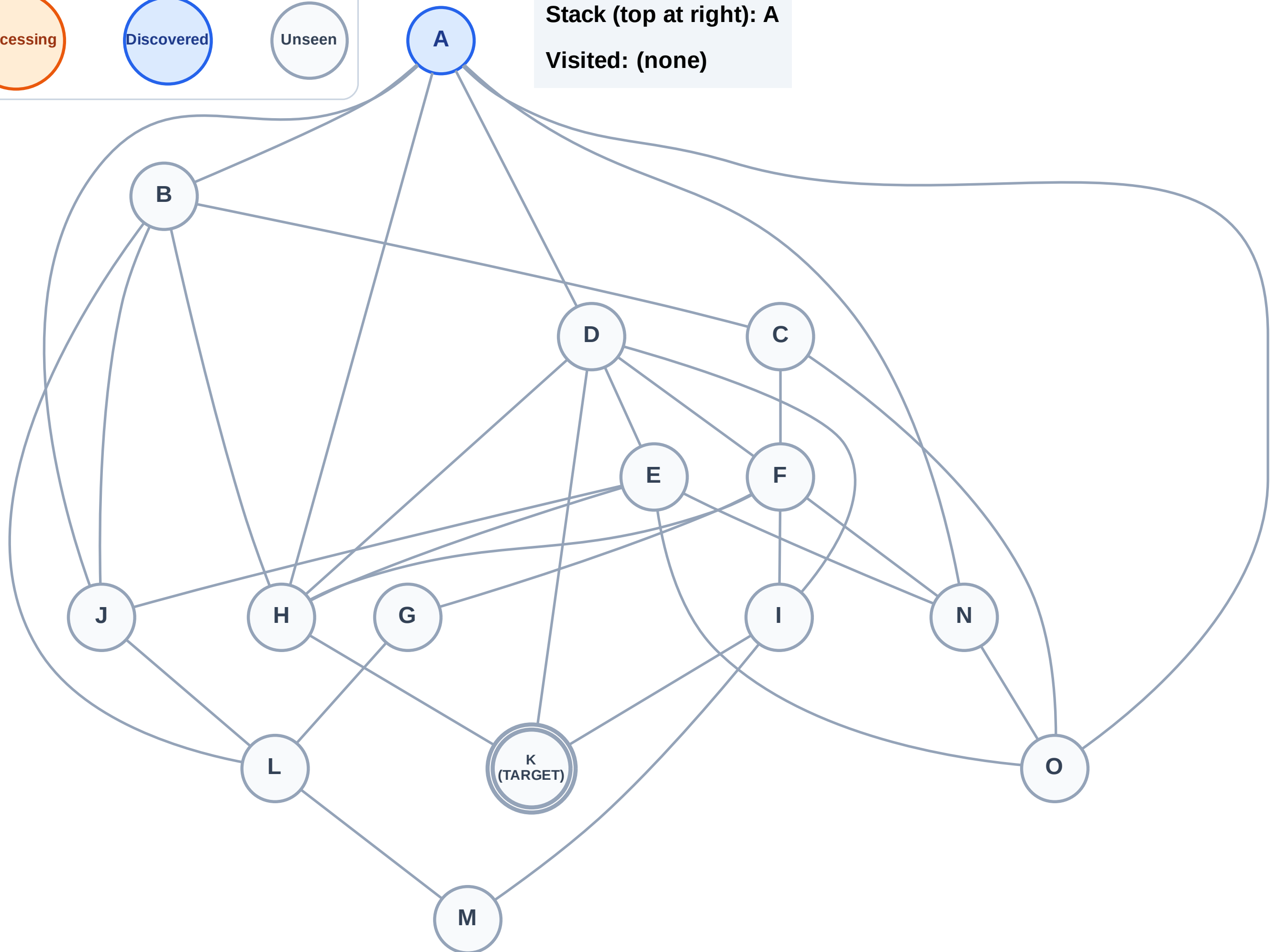


DFS Traversal

Step 1: Start at A (Seeking target node K)

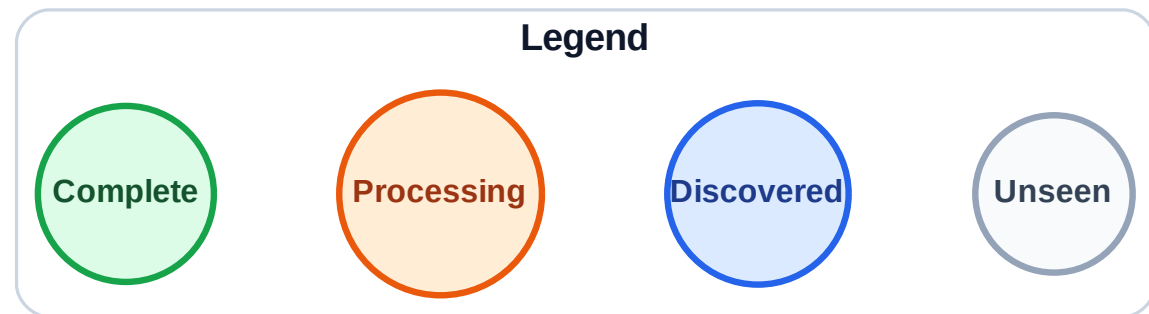


Stack (top at right): A
Visited: (none)

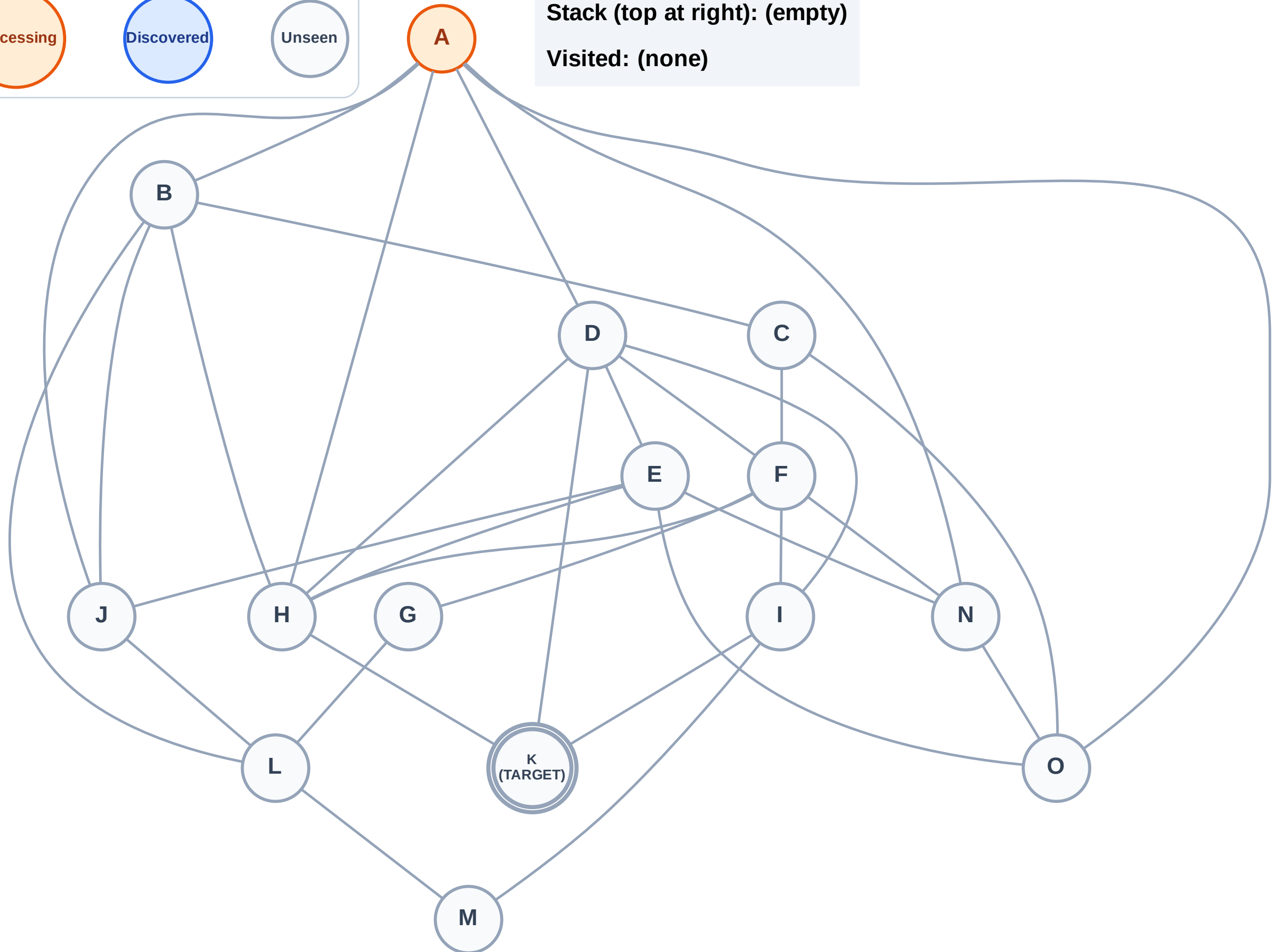


DFS Traversal

Step 2: Process node A



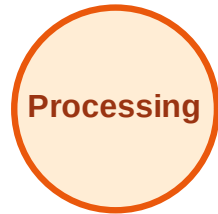
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

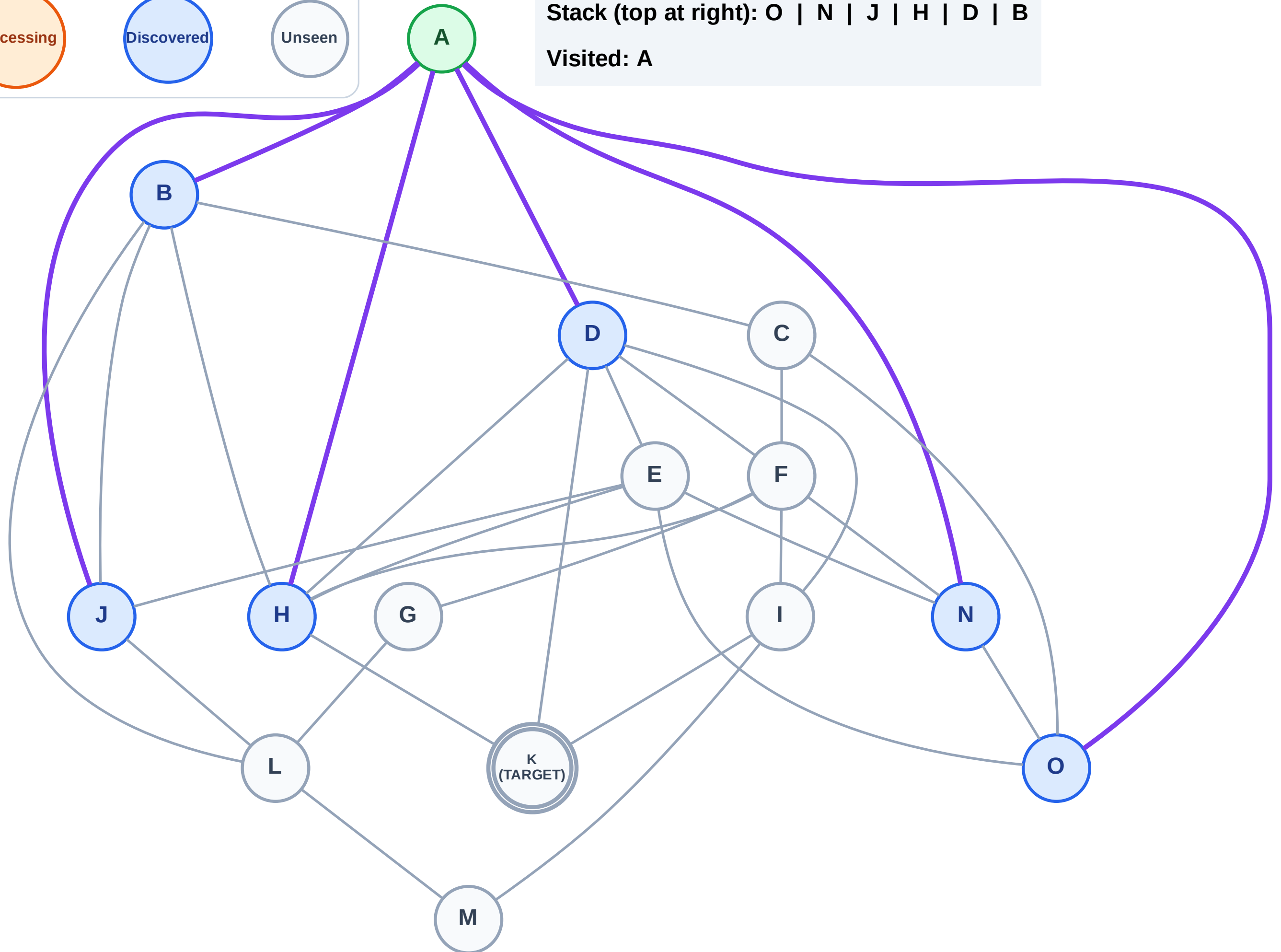
Step 3: Discover O, N, J, H, D, B from A

Legend



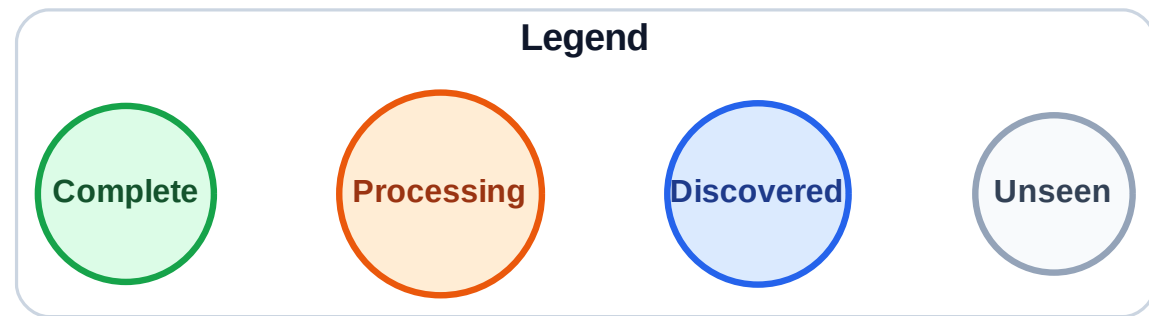
Stack (top at right): O | N | J | H | D | B

Visited: A



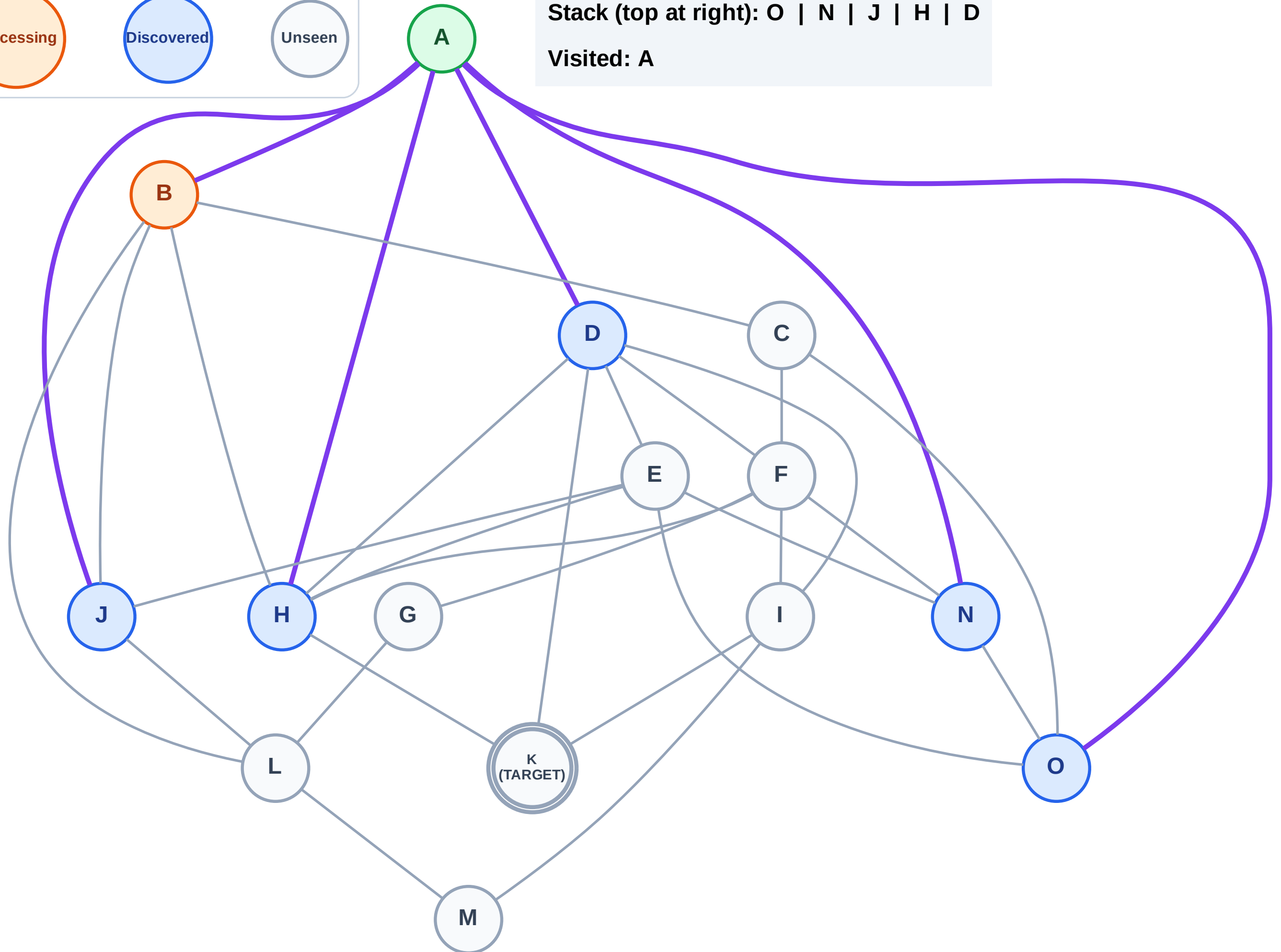
DFS Traversal

Step 4: Process node B



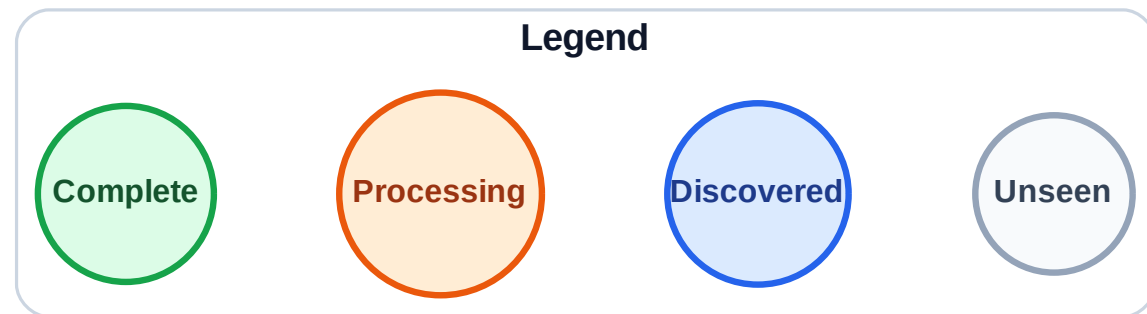
Stack (top at right): O | N | J | H | D

Visited: A



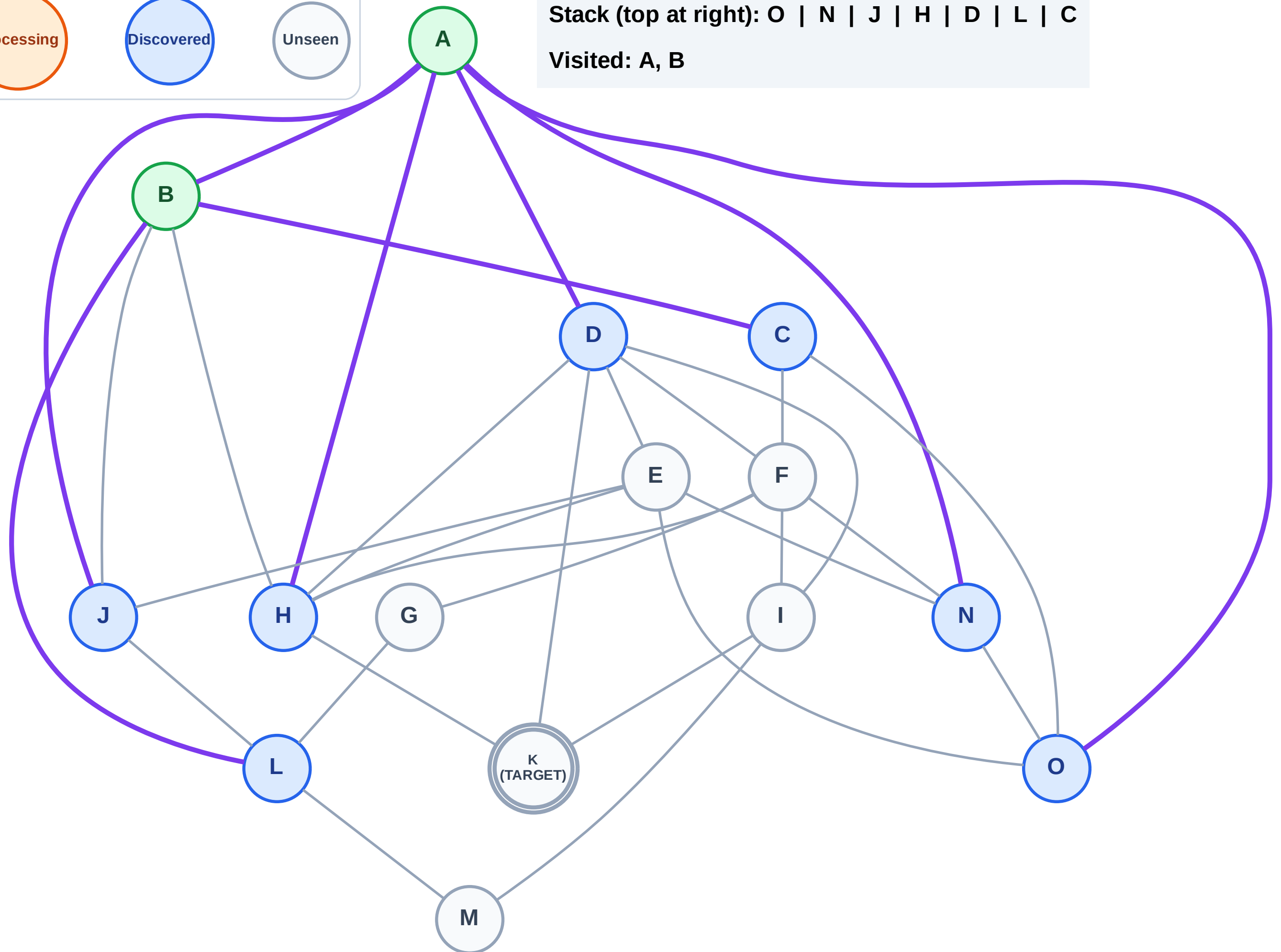
DFS Traversal

Step 5: Discover L, C from B



Stack (top at right): O | N | J | H | D | L | C

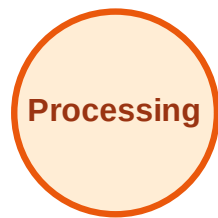
Visited: A, B



DFS Traversal

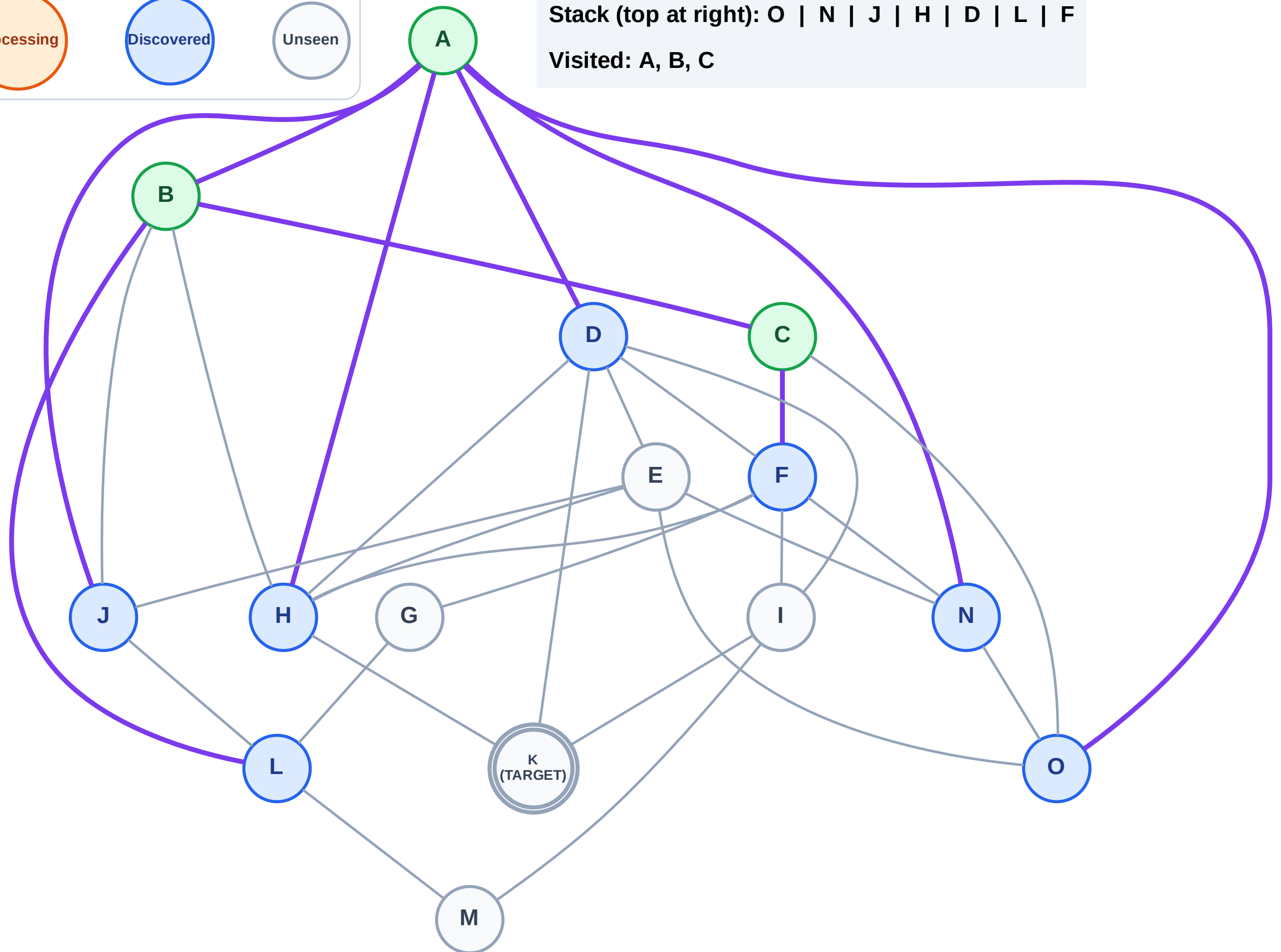
Step 7: Discover F from C

Legend



Stack (top at right): O | N | J | H | D | L | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

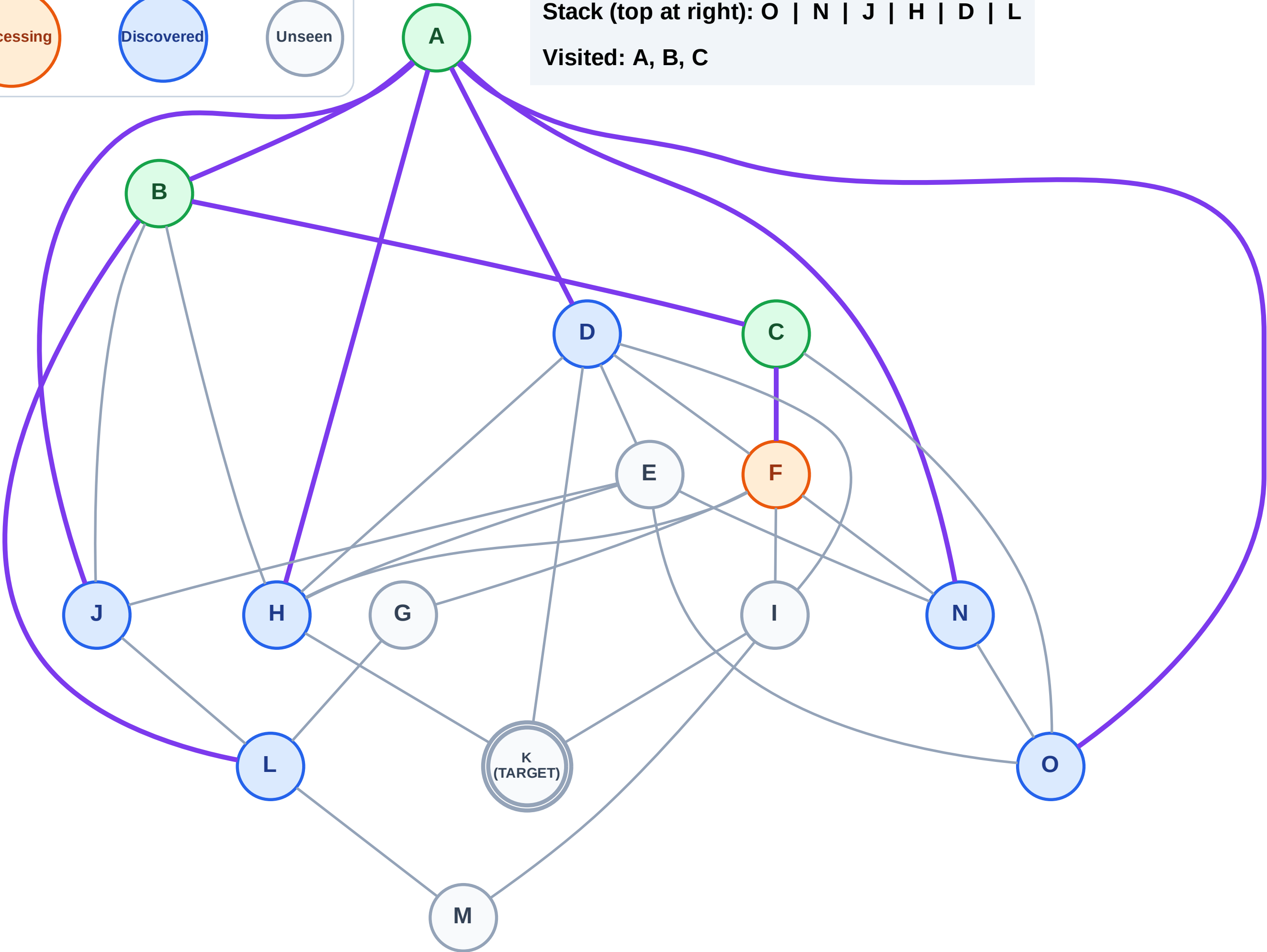
Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L

Visited: A, B, C



DFS Traversal

Step 9: Discover I, G from F

Legend

Complete

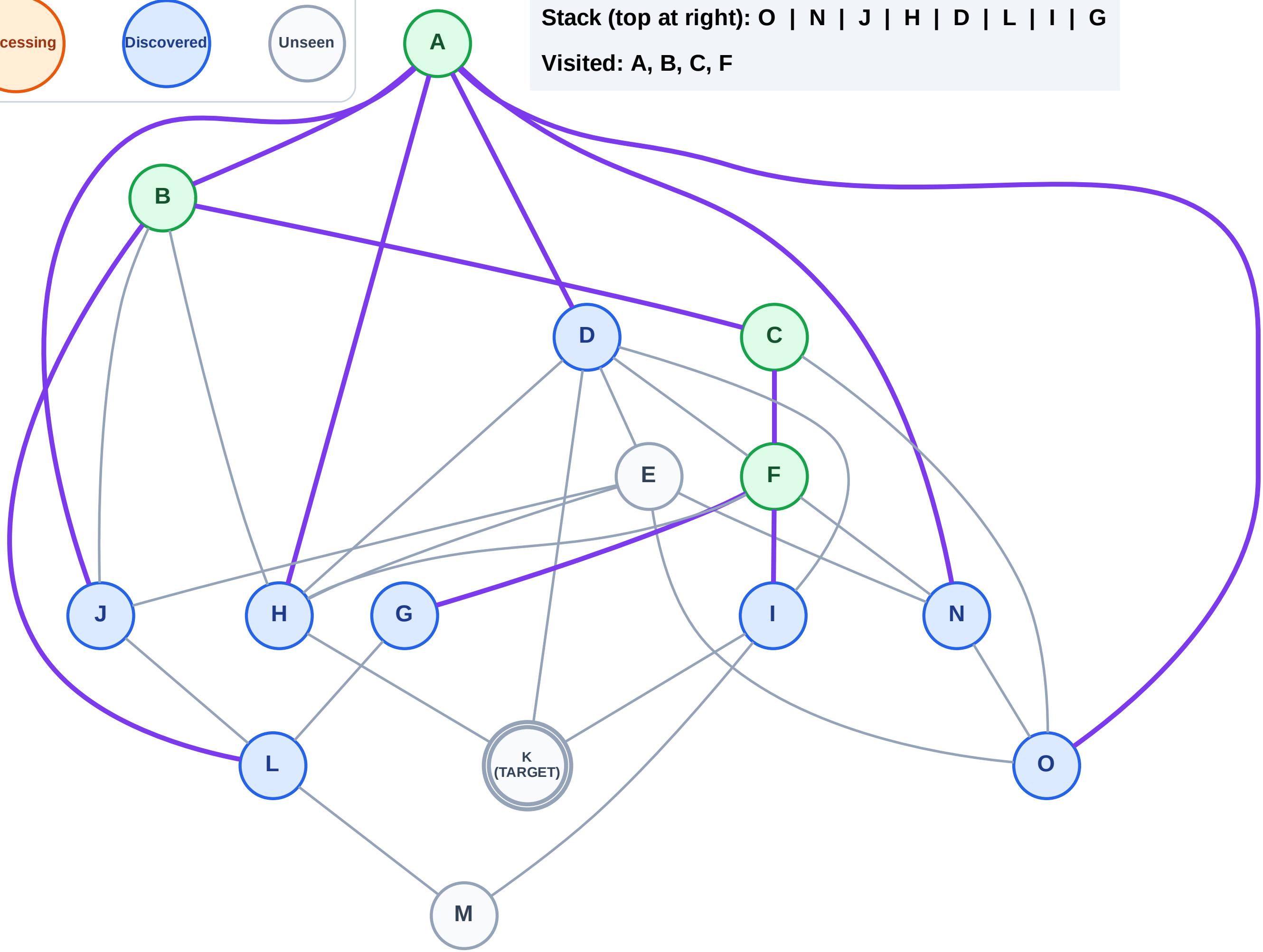
Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L | I | G

Visited: A, B, C, F



DFS Traversal

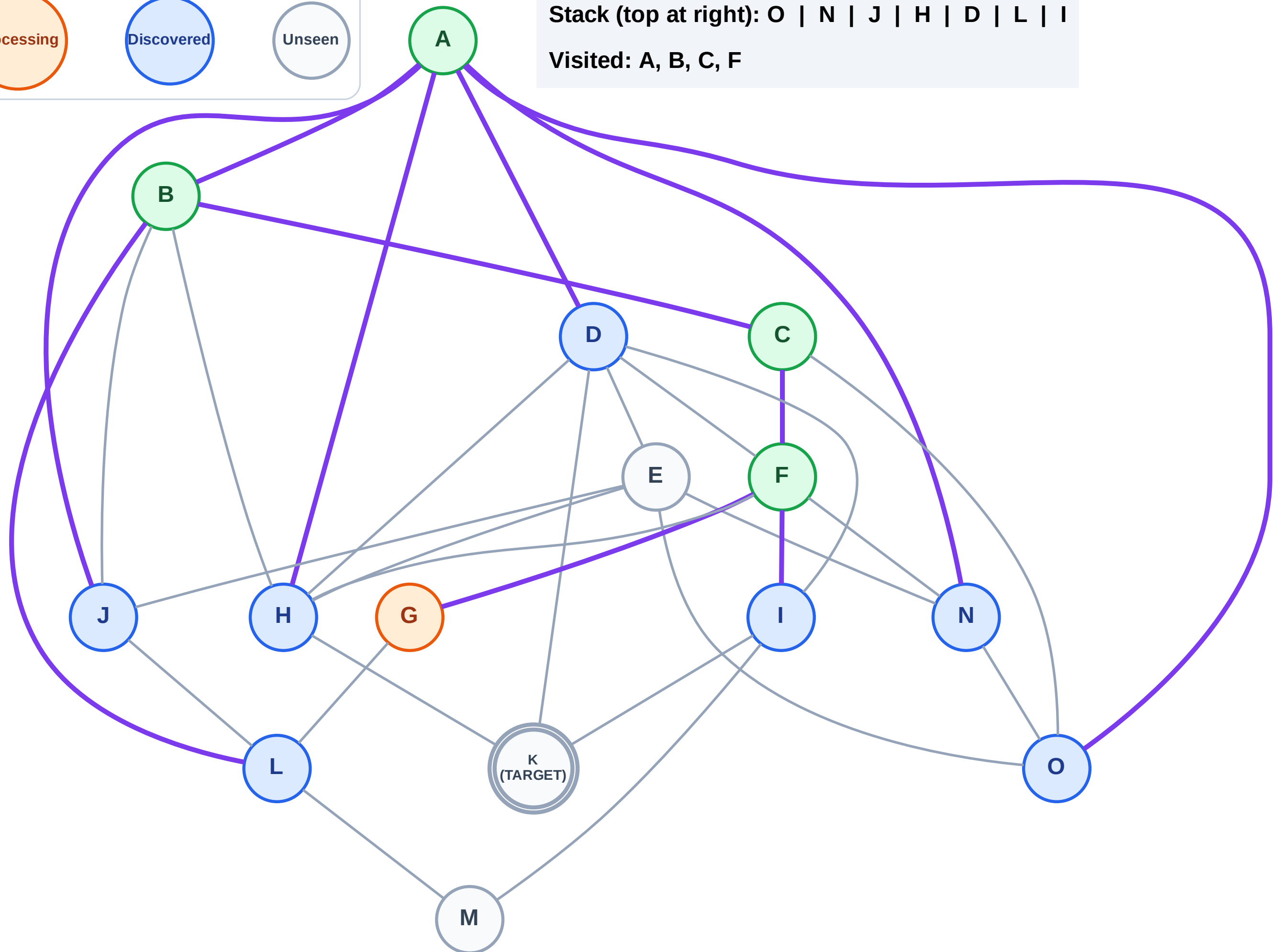
Step 10: Process node G

Legend

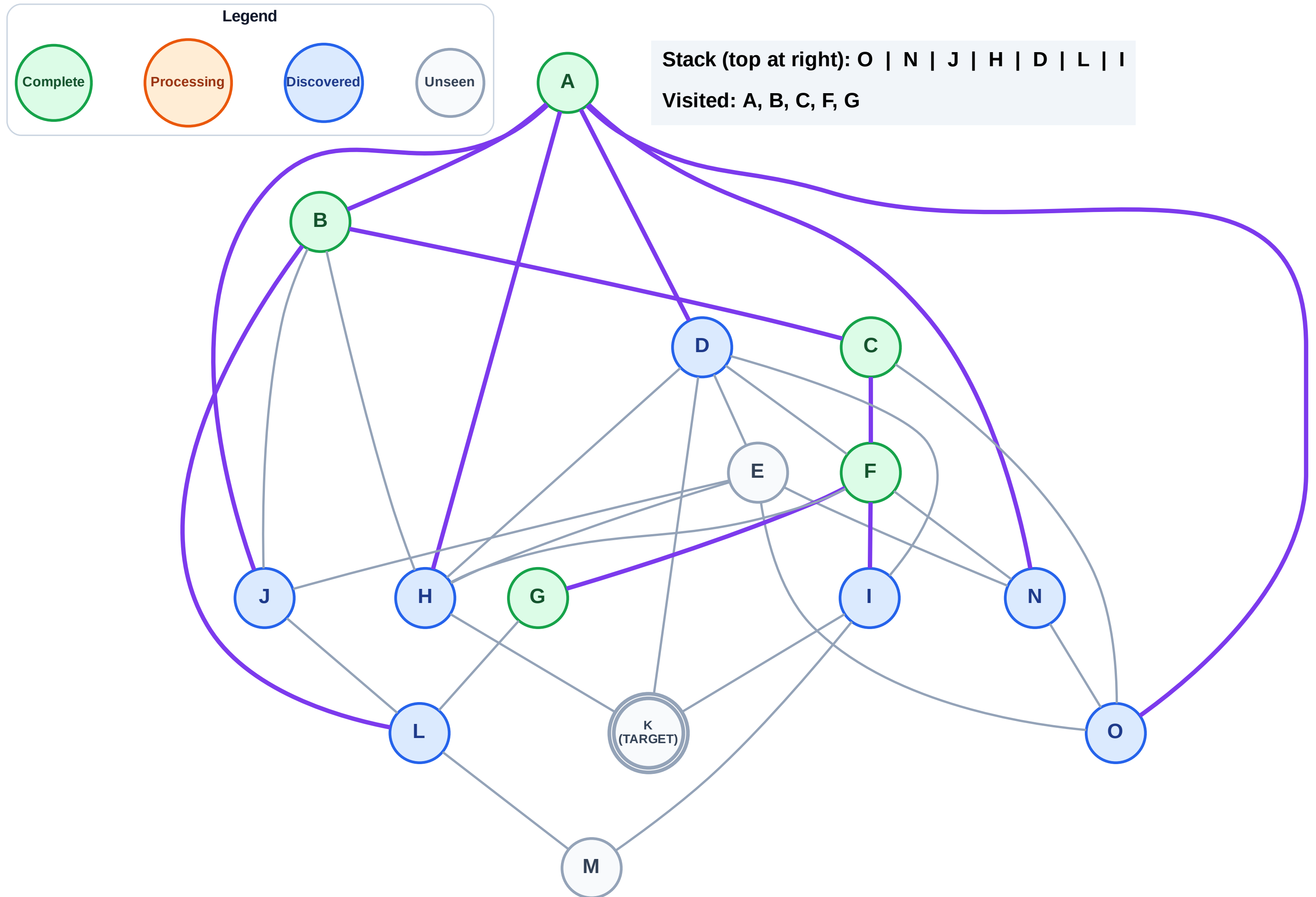


Stack (top at right): O | N | J | H | D | L | I

Visited: A, B, C, F

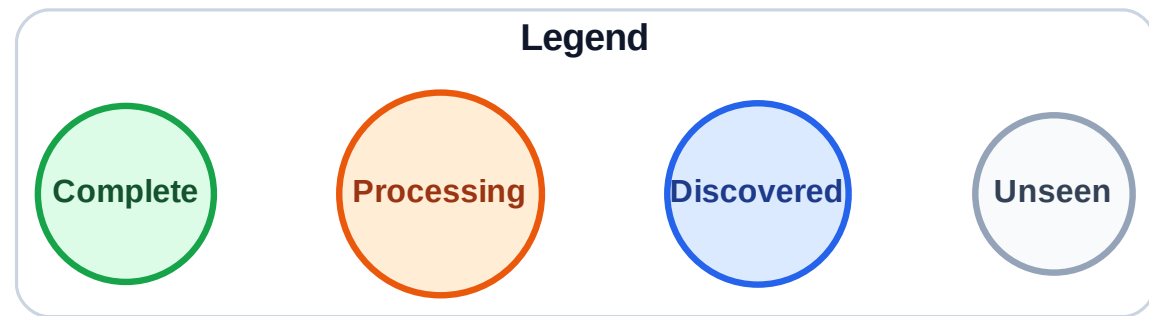


Step 11: No new neighbors from G



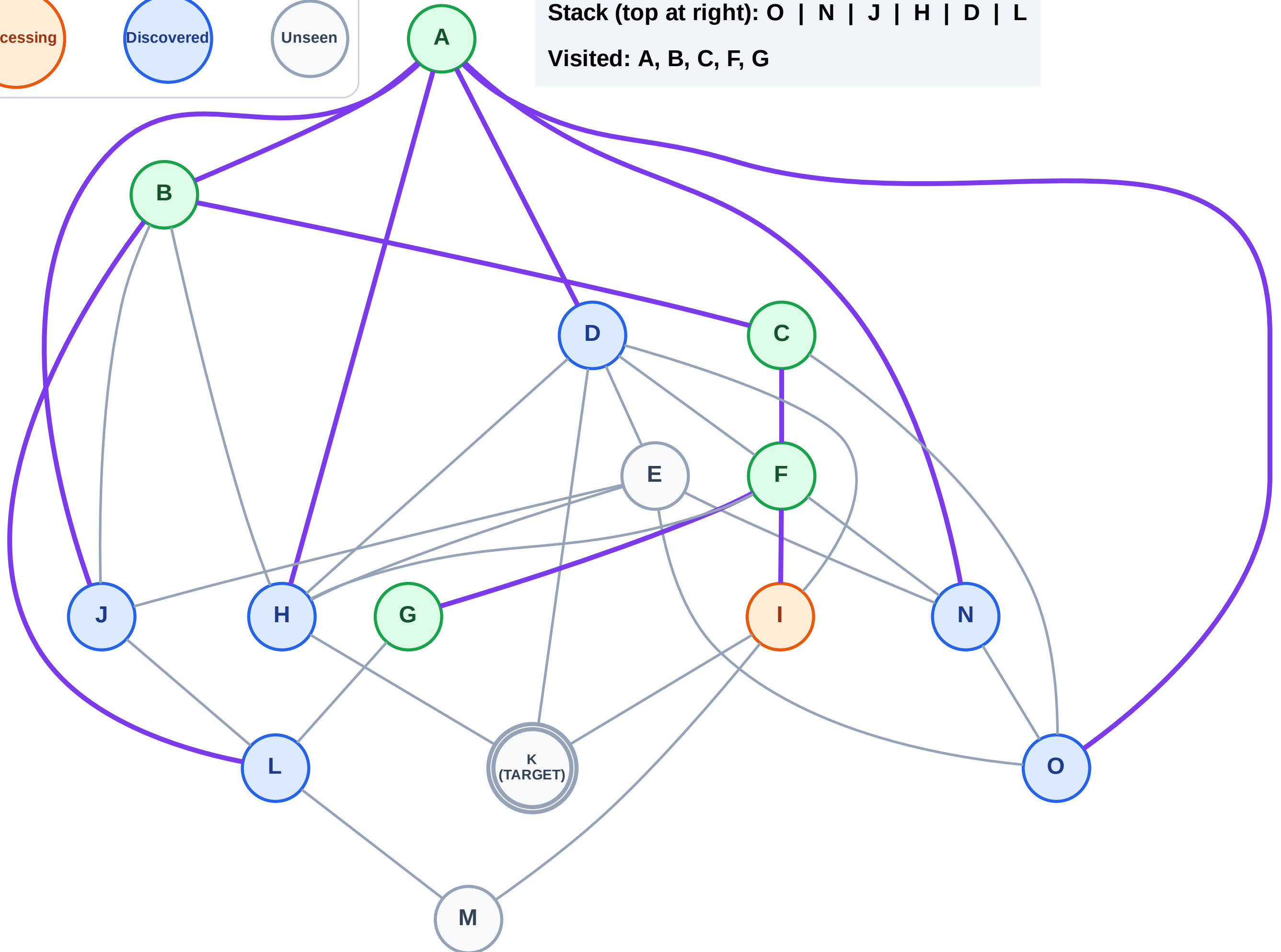
DFS Traversal

Step 12: Process node I



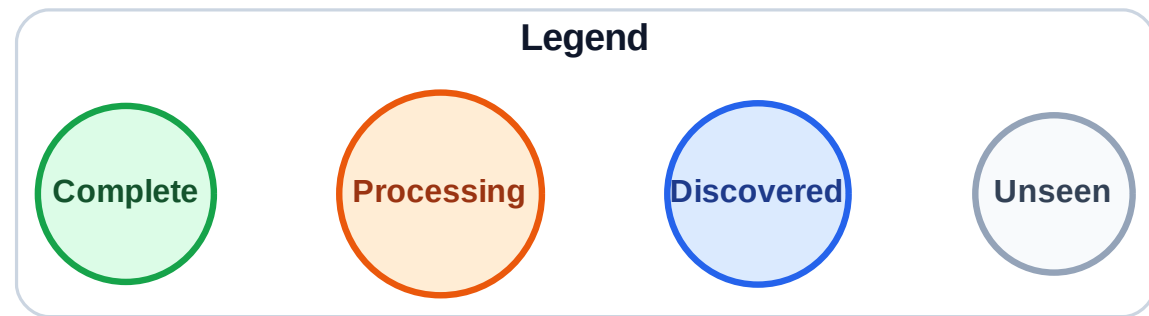
Stack (top at right): O | N | J | H | D | L

Visited: A, B, C, F, G



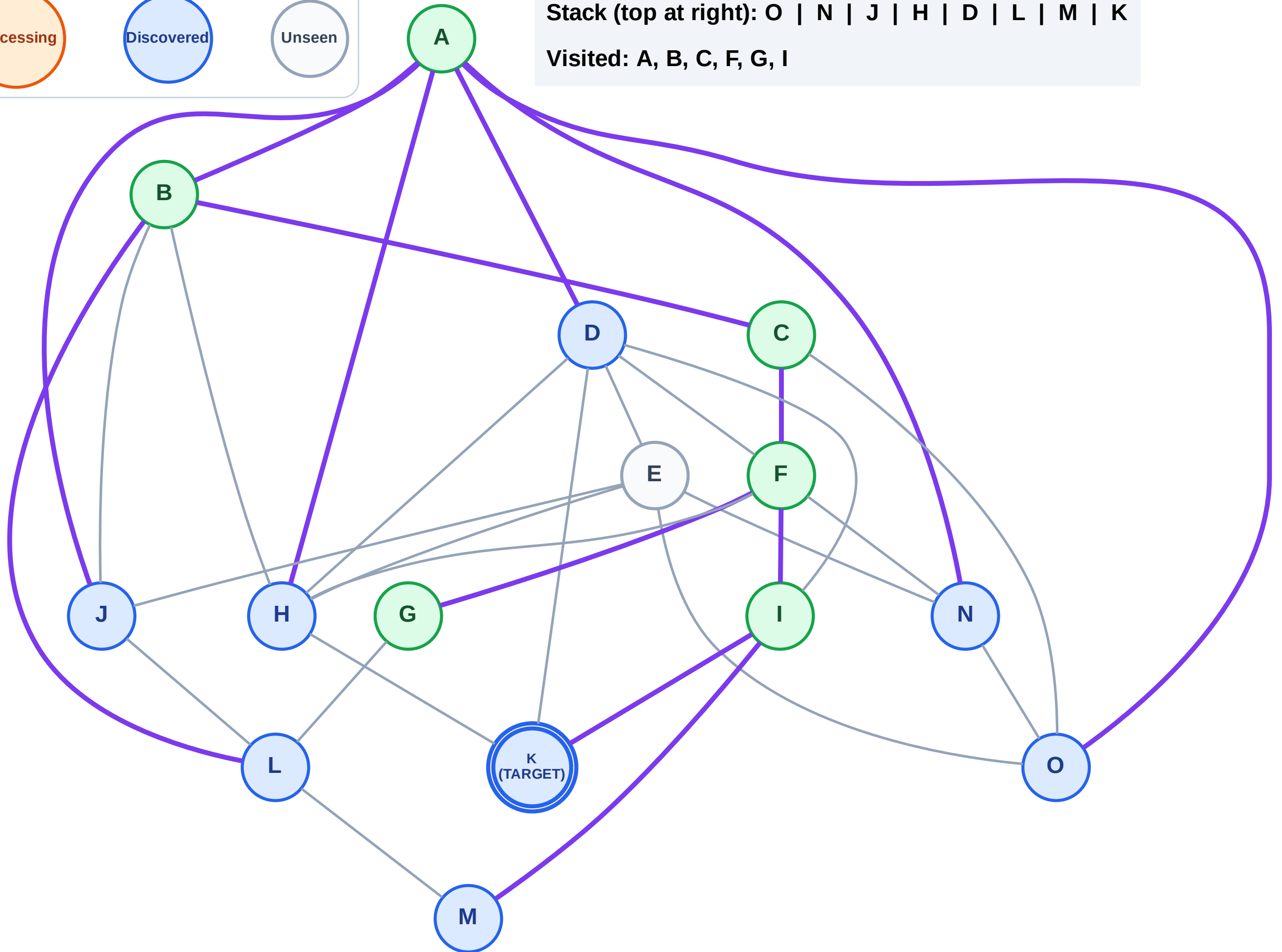
DFS Traversal

Step 13: Discover M, K from I



Stack (top at right): O | N | J | H | D | L | M | K

Visited: A, B, C, F, G, I



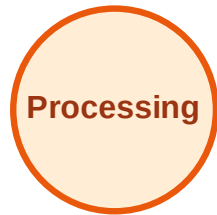
DFS Traversal

Step 14: Process node K

Legend



Complete



Processing



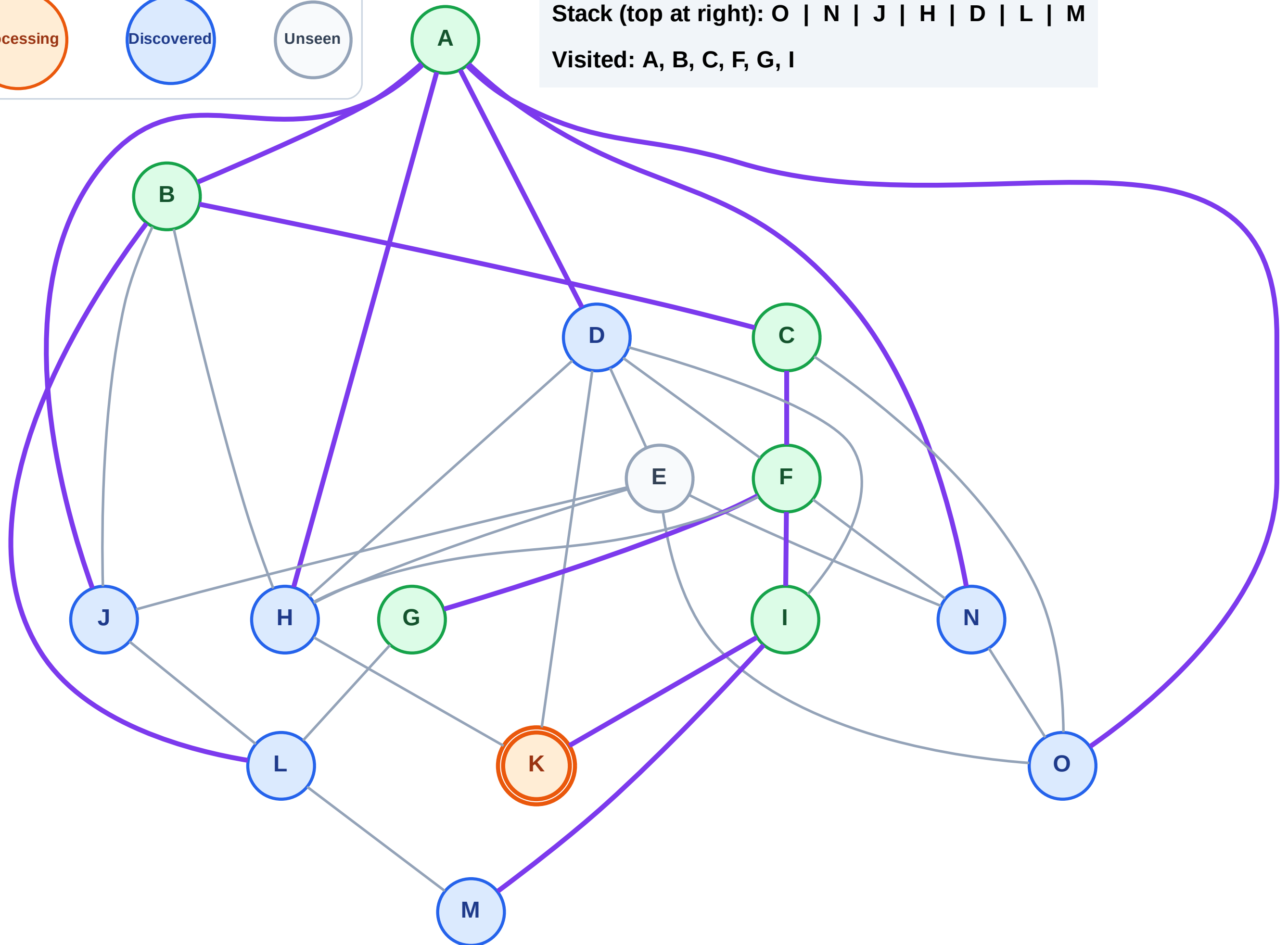
Discovered



Unseen

Stack (top at right): O | N | J | H | D | L | M

Visited: A, B, C, F, G, I



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (5 hops): A → B → C → F → I → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | J | H | D | L | M

Visited: A, B, C, F, G, I, K

